

# QuarkyLab

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## Comprehensive Infrastructure Assurance Review

Evidence-based reliability, security, recoverability, and enterprise-alignment assessment of the NetFRAME / km-cluster home lab

|                       |                                                            |
|-----------------------|------------------------------------------------------------|
| <b>Environment</b>    | NetFRAME / km-cluster (7-node Proxmox VE 9.2.3)            |
| <b>Operator</b>       | Kyle Mason (machismo0311), Greater Cleveland OH            |
| <b>Review date</b>    | 2026-07-23                                                 |
| <b>Mode</b>           | Read-only. No configuration changes made.                  |
| <b>Basis</b>          | Live SSH evidence (2026-07-23) + repo audit commit 65f3681 |
| <b>Classification</b> | Internal - contains topology; no secrets reproduced        |

Prepared as a read-only assurance review. Every finding is tied to environment evidence. No system was restarted, reconfigured, or modified. All change proposals are held for explicit operator approval.

<Home-Lab/docs/audits/quarkylab-comprehensive-review/>

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# 1 Executive Summary

## One-paragraph verdict

QuarkyLab, the NetFRAME / km-cluster home lab, is an unusually mature environment that runs three genuinely production workloads at once: DUNE neutrino-physics GPU research, a multi-tenant student AI/ML teaching cluster, and a self-hosted LLM inference platform. Live read-only verification on 2026-07-23 found the fabric healthy: a 7 of 7 quorate Proxmox cluster, GPUs correctly assigned, backups fresh (14 snapshots dated 2026-07-23), both UPS units online at full charge, and the tracked repositories secrets-clean. The environment is well ahead of typical home-lab practice in operational discipline. The material risks are concentrated, well understood, and mostly already in the operator's own backlog: a single faulted storage drive on a pool that has no second copy, a backup topology that is effectively one-copy / one-site, and two single points of failure (OPNsense and Randy) that the operator is already planning to address.

## 1.1 Overall assessment

This is a strong environment. The header-line facts were independently confirmed against live hardware rather than taken from documentation (E-01 through E-08). The cluster is quorate with all corosync links connected (E-01); QuarkyLab carries one RTX 8000 (48 GB) and Jarvis two RTX 6000 (24 GB each), which confirms the repository and *contradicts* the review specification's assertion that the assignment was reversed (E-02, E-03, DR-01); the ZFS datastore pool is clean and last night's backups completed (E-06, E-08). Nothing found during this review indicates an imminent outage. The weaknesses are real but bounded, and the operator has already produced a self-audit (commit 65f3681, 2026-07-22) that names most of them.

## 1.2 Major strengths

- **Deterministic AI-operations safety.** The netframe-monitor / Jarvis stack is genuinely enterprise-grade in its safety design: a single deterministic evidence engine that every report path uses, model self-confidence stripped and forbidden, a three-action human-gated remediation allowlist with no autonomous execution, universal policy screening including the chat and Discord surfaces, untrusted-telemetry handling, and full survivability if the model is offline (E-14).
- **Disciplined runbook and disaster-recovery culture.** Formal RCAs and AARs, ADRs, an offline break-glass credential path independent of the cluster, a node-unreachable triage runbook, OPNsense serial-console DR with age-encrypted config backups, and a laptop-recovery runbook (E-21).
- **Real, applied network segmentation.** Seven VLANs, BMCs isolated to VLAN 20 with an *applied* egress clamp, a VLAN 30 to VLAN 1 block, and DNS pinholes for isolated VLANs (E-21). This is deployed, not aspirational.
- **Config-as-code monitoring.** Grafana alerting, exporters, and textfile collectors are versioned; there are dead-man heartbeats for the pipeline itself (E-15).
- **Honest self-audit already in flight.** The operator is mid-way through a reconciliation branch that catalogs the documentation drift found here.

## 1.3 Highest risks

Ordering below reflects the corrections established late in the review. The storage items were downgraded on live evidence, and the alerting item was root-caused and closed.

1. **HIGH Backups are effectively one-copy / one-site (QL-R03).** Every backup lives only on Randy: one PBS, one ZFS pool, one rack, one UPS bus (E-08). This is now the single largest unmitigated exposure.
2. **HIGH OPNsense is a single point of failure (QL-R04).** One VM on pve2 provides routing, firewall, DHCP, and inter-VLAN gateway for the whole LAN.
3. **HIGH Randy is a single storage node (QL-R05).** PBS, research NFS, and RKE2 persistent volumes all depend on it.
4. **MEDIUM A live GitHub token sits in plaintext at rest (QL-R07).** ~/github\_helper.py contains a classic PAT; it is gitignored and never committed, but usable if the workstation is compromised (E-17). Two other plaintext secret files sit alongside it.
5. **MEDIUM Reduced parity on bulk (QL-R01), impact corrected.** mpathv is FAULTED in raidz2-2, leaving one parity drive. Originally rated urgent on the assumption that bulk held DUNE research data; live inspection (E-29) shows the pool is essentially empty, so this is loss of redundancy rather than imminent data loss. Replacement is pending operator hardware.

6. **MEDIUM** **bulk still has no second copy (QL-R02).** Low impact today because the pool is near-empty, but the exposure becomes real the moment research data migrates onto it, so the second copy must exist first. The Ares restic backup also currently targets this degraded pool.
7. **MEDIUM** **Documentation drift (QL-R19).** Recent changes landed in CLAUDE.md but not in README, topology, or the docs mirror; two review-spec "corrections" are inverted versus live reality and must not be applied. This review added a third: a phantom EX2300 switch that has never existed (DR-15).

## 1.4 Changes applied during this review

The review was read-only by default. Two corrective changes were made after explicit operator approval, and both were verified afterwards by the reviewer.

- **ZFS device-level alerting deployed (QL-R11, closed).** The ZfsPoolDegraded rule could never have fired for bulk: it watches a pool-level health metric that reads ONLINE while raidz2 retains redundancy (max\_over\_time over 7 days = 0). No per-device metric existed. A new node\_zfs\_pool\_nonhealthy\_devices metric and a critical ZfsDeviceFaulted rule were added and deployed to Randy, QuarkyLab and Jarvis; verified live in Prometheus with bulk = 1 and all other pools 0.
- **EX3400 pinned as spanning-tree root (QL-R23, closed).** No bridge-priority was configured, so a UniFi access switch had won the root election by MAC address while documentation claimed the core was root. bridge-priority 4096 was committed; the core is now genuinely root.
- **Not done: the drive replacement (QL-REC01)** remains pending operator hardware.

## 1.5 Dimension assessments

- **Reliability:** Strong at the compute/cluster layer (7/7 quorum, pinned GPU stacks intact); weakened by concentrated SPOFs (OPNsense, Randy) and HA fencing left in standby.
- **Security:** Well above home-lab norm - applied segmentation, least- privilege API tokens, working SIEM. Residual: at-rest workstation secrets, unauthenticated LAN AI/registry endpoints, guest SIEM blind spot.
- **Recoverability:** Good documentation and an offline break-glass path; limited by the single backup site and one unverified restore-readiness gap (pve1 unbacked).
- **Power resilience:** Both UPS online and healthy, but UPS-A is loaded to 59 percent with roughly 9.5 minutes of runtime (E-07), an imbalance worth rebalancing.
- **AI-system reliability:** Excellent safety posture; the remaining lift is calibration/eval, not trustworthiness.
- **Scientific and student workloads:** Genuinely well engineered - SLURM shard plus per-job MPS VRAM caps plus aptainer isolation, research preemption, and reproducible container images.
- **Enterprise alignment:** Comparable-to-mature in monitoring, docs, IaC, and secrets discipline; material gaps in HA/redundancy and offsite DR.

## 1.6 Top 10 prioritized recommendations

1. Replace the faulted bulk drive and scrub (QL-REC01, urgent).
2. Verify the ZfsPoolDegraded alert actually reached Discord (QL-REC02, read-only).
3. Add a second backup copy / restore the 3-2-1 principle (QL-REC03).
4. Back up the standalone pve1 guests (QL-REC04).
5. Rotate the at-rest GitHub PAT and the two plaintext secret files (QL-REC05).
6. Deploy prometheus-pve-exporter for cluster/quorum/guest metrics (QL-REC06).
7. Add the missing alert rules: PBS-stale, verify-failed, broad cert-expiry (QL-REC07).
8. Rebalance UPS loads and confirm NUT graceful-shutdown sequencing (QL-REC08).
9. Reconcile EX3400 SSH access and capture switch config to git via Oxidized (QL-REC09).
10. Land the documentation reconciliation PR (D-01 through D-10) (QL-REC11).

Building the OPNsense CARP HA pair (QL-REC10) is the highest-value medium-term structural improvement and removes the top availability SPOF.

## 1.7 Explicitly accepted risks and where no change is recommended

The following are sound decisions that should stand: the pinned 6.14.11-9-pve kernels on the GPU nodes must not be upgraded (QL-REC20); the raidz1 vdevs on QuarkyLab and Jarvis are an accepted single-parity tradeoff given hot spares, clean scrubs, and low utilization (QL-R21); single-node observability is a reasonable posture at this scale given the dead-man

heartbeats (QL-R10); and the VLAN 1 egress lockdown remains deliberately unenforced pending its prerequisites. Stable systems should not be changed without a compelling reason, and most of this environment is stable.

## 1.8 Confidence and limitations

Confidence in the live-verified conclusions (cluster health, GPU assignment, storage state, backup freshness, power state, and now the switch) is **high**: each was confirmed by direct read-only query and cross-checked against the prior audit. The EX3400 was inspected live later in the review once the operator supplied credentials (E-22, E-25), which lifted the original switch limitation and produced the spanning-tree finding below. Confidence in the firewall control plane remains **medium**: OPNsense was not queried (no credentials used, by design, E-23), so firewall specifics still rest on repository documentation. The review was read-only by default; the two changes listed below were made only after explicit operator approval, and were verified afterwards. See Section 21 for the full limitations register.

## 2 Scope and Methodology

### 2.1 Scope

This review evaluated the QuarkyLab / NetFRAME home lab across 21 assessment dimensions: reliability, availability, recoverability, security, network correctness, configuration consistency, monitoring quality, alert quality, storage integrity, backup quality, power resilience, energy efficiency, capacity planning, operational maturity, documentation quality, automation safety, AI-system reliability, scientific-workload support, student-environment isolation, recruiter and hiring-manager presentation, and Fortune-100 enterprise alignment.

In scope were only systems, repositories, and services that belong to the home lab: the 7-node km-cluster plus the standalone pve1 host, Randy and the DS4246 shelf, the Juniper and UniFi switching, OPNsense, Headscale, the RKE2 cluster, the Docker/LXC service estate, the Ollama/Open WebUI/llm\_router AI stack, the Grafana and Wazuh observability planes, the UPS/PDU/power layer, and the operator's GitHub repositories. No unrelated external system was probed. External services already used by the environment (package repositories, GitHub, container registries, certificate authorities, DNS resolvers) were treated as passive dependencies, not targets.

### 2.2 Operating posture: read-only by classification

Every command was classified before execution as read-only/low-risk, read-only/resource-intensive, configuration-changing, service-affecting, destructive, or external-target interaction. Only read-only/low-risk commands were run without approval. No configuration-changing, service-affecting, destructive, or external-interaction command was executed. Concretely, this review performed no restarts, reboots, reloads, migrations, deletions, rotations, or deployments; used no discovered credential; injected no failures; and did not enforce the VLAN 1 egress lockdown, in line with the standing decision that its prerequisites are not yet met. Where confidence that an action was non-disruptive fell below the required threshold, the action was not taken and the limitation was recorded (see Section 21).

### 2.3 Evidence approach

The review compared five state layers where possible: declared (the review specification and prior intent), configured (files and definitions), deployed (running processes and guests), observed (live telemetry), and documented (the repository). Documentation was not treated as true merely because it exists, and live output was not treated as authoritative where it represents only transient runtime state.

A key input was the operator's own prior audit (commit 65f3681, 2026-07-22), which had already captured per-node live evidence and built a canonical fact ledger. That corpus is one day old and was used as generated evidence to be re-verified, not trusted blindly. This review independently re-ran the highest-stakes checks against current live hardware.

### 2.4 Independent validation (multi-method)

For every high-severity conclusion, at least two independent read-only methods were used. Examples actually applied in this review:

- **GPU assignment**: live `nvidia-smi` on both R730s (E-02, E-03) versus the review specification and the fact ledger. The live reading settled the contradiction in favor of the repository (DR-01).
- **Storage health**: live `zpool status` on Randy (E-04) versus the ledger's canonical geometry and the prior audit's finding on `mpathv`.

- **Cluster and guests:** live `pvecm status` plus `pvesh get /cluster/resources` (E-01, E-06) versus the documented node and service placement, which surfaced the Grafana/Headscale relocation drift.
- **Power:** live `upsc` readings (E-07) versus the reliability assessment’s UPS-imbalance finding, which confirmed UPS-A is the loaded unit.
- **Backups:** live PBS snapshot listing dated 2026-07-23 (E-08) versus the documented backup schedule.

## 2.5 What was not done

No failure injection, no restore into production, no restarts, and no live query of OPNsense (E-23). The EX3400 exclusion was lifted mid-review when the operator supplied credentials, so switch findings are live-verified (E-22, E-25).

Two configuration changes were made, both *only* after explicit operator approval and both verified afterwards: the ZFS device-level metric and alert rule, and the EX3400 bridge-priority. Everything else remained read-only. Findings that still rest on documentation are marked medium-confidence and carry a precise follow-up request.

## 2.6 Deliverable structure

The deliverables are this LaTeX report (sections plus appendices), a set of machine-readable registers under `data/` (evidence, assets, services, drift, risk, recommendations, backup matrix, alert matrix), and text-based topology diagrams under `diagrams/` (physical, logical, service-dependency, recovery-dependency, power). The `evidence/` directory describes evidence locations and collection methods rather than copying sensitive output.

## 3 Evidence Register

Every finding in this review traces to a numbered evidence item. The register below records the source, the system it came from, the exact command or path, its age, a reliability tier, and the limitations that bound how far it can be trusted. The machine-readable form is `data/evidence-register.csv`.

### 3.1 Reliability tiers and evidence types

Evidence is typed as *live* (a direct read-only query executed during this review, 2026-07-23), *live-captured* (the prior audit’s node captures, one day old), *generated* (assessment reports produced by prior tooling), *inferred* (conclusions read out of source code or configuration), or *manual* (operator-maintained documentation). Live evidence is weighted highest for current-state claims; documented evidence is weighted highest for intent and history. Where the two conflict, the conflict itself is recorded rather than silently resolved: the most consequential examples are the GPU assignment (live overrides the specification, DR-01), the OPNsense version (documented as 25.1.12, not re-verified live, E-23 / DR-02), and Randy’s PVE version, where the live reading of 9.2.3 supersedes the documented 9.1.1 (E-04 / DR-10).

### 3.2 Register

| ID   | Source / system       | Path or command                                                    | Age / rel.    | Type / limitation                               |
|------|-----------------------|--------------------------------------------------------------------|---------------|-------------------------------------------------|
| E-01 | Live SSH / pve3       | <code>pvecm status; corosync-cfgtool -s</code>                     | live / high   | live; transient runtime state                   |
| E-02 | Live SSH / quarkylab  | <code>nvidia-smi; zpool list; qm list; sinfo</code>                | live / high   | live; confirms GPU vs stale spec                |
| E-03 | Live SSH / jarvis     | <code>nvidia-smi; zpool list; systemctl is-active</code>           | live / high   | live; confirms 2x RTX 6000                      |
| E-04 | Live SSH / randy      | <code>zpool status bulk/datastore; pbs datastore list</code>       | live / high   | live; <code>multipath -ll</code> returned empty |
| E-05 | Live SSH / pve1       | <code>pveversion; pct list; apt list --upgradable</code>           | live / high   | live; 142 pending vs audit 133                  |
| E-06 | Live SSH / pve3       | <code>pvesh get /cluster/resources; storage.cfg; ha-manager</code> | live / high   | live; cluster-wide snapshot                     |
| E-07 | Live SSH / pve3       | <code>upsc triplite; upsc midatlantic</code>                       | live / high   | live; instantaneous load only                   |
| E-08 | Live SSH / randy      | <code>ls -dt /datastore/{ct,vm,host}/*/2026-07-23</code>           | live / high   | live; filesystem listing                        |
| E-09 | Repo audit / Home-Lab | <code>audit/findings/*.md</code> (commit 65f3681)                  | 1 day / high  | generated; two spec corrections inverted        |
| E-10 | Repo audit / Home-Lab | <code>audit/live/*.txt</code>                                      | 1 day / high  | live-captured; superseded where re-run          |
| E-11 | Repo audit / Home-Lab | <code>audit/facts/ledger.yaml</code>                               | 1 day / high  | inferred; canonical-vs-stale table              |
| E-12 | Repo / Home-Lab       | <code>CLAUDE.md</code>                                             | 2 days / high | manual; most-accurate doc                       |
| E-13 | Repo / Home-Lab       | <code>README.md; topology/; vault MOC</code>                       | 1 day / med   | manual; lags CLAUDE.md                          |



| ID   | Source / system                | Path or command                                    | Age / rel.    | Type / limitation                                    |
|------|--------------------------------|----------------------------------------------------|---------------|------------------------------------------------------|
| E-14 | Code read / netframe-monitor   | *.py; constitution/; eval/                         | live / high   | inferred; static read not runtime trace              |
| E-15 | Code read / monitoring-stack   | grafana-provisioning/alerting/*.yaml               | 1 day / high  | inferred; rules not eval-traced                      |
| E-16 | Code read / netframe-dashboard | serve.py; README.md                                | 1 day / high  | inferred; some sources fall back to mock             |
| E-17 | Secrets sweep / home dir       | grep + git ls-files across repos                   | live / high   | live; PAT at rest in github_helper.py                |
| E-18 | Prior assessment / enterprise  | report/*.tex; risk-register.tex                    | 10 days / med | generated; some items since resolved                 |
| E-19 | Prior assessment / security    | report/*.tex; MASTER-TODO.md                       | live / high   | generated; reconciled 2026-07-22                     |
| E-20 | Prior assessment / reliability | report/*.tex                                       | 9 days / med  | generated; windowed items pending                    |
| E-21 | Repo docs / Home-Lab           | vault/Runbook/*.md (AARs, DR, CARP)                | live / high   | manual; operator narrative                           |
| E-22 | Live SSH / ex3400              | show version/chassis/spanning-tree/vlans/          | live / high   | live; key auth denied, password supplied by operator |
| E-25 | Live SSH / ex3400              | show configuration   display set (241 lines)       | live / high   | live; no bridge-priority; confirms D-08              |
| E-26 | Live SSH / 3 ZFS nodes         | zpool status parsing validation                    | live / high   | live; bulk=1, datastore=0                            |
| E-27 | Live query / Prometheus        | node_zfs_pool_nonhealthy_devices; 7d max_over_time | live / high   | live; proves rule could never fire                   |
| E-28 | Live SSH / ex3400              | post-change show spanning-tree bridge              | live / high   | live; Root ID == Bridge ID == 4096                   |
| E-29 | Live read / randy              | zfs list -r bulk                                   | live / high   | live; bulk near-empty, corrects QL-R01/R02           |
| E-23 | Not collected / opnsense       | no API creds used; VM 100 untouched                | n/a           | not-collected; 25.1.12 per AAR only                  |
| E-24 | Repo / ansible playbooks       | roles/{hardening,backup_verify}                    | 1 day / high  | manual; --check only, not applied                    |

### 3.3 Notable conflicts surfaced

The register makes three conflicts explicit. First, the review specification's GPU "correction" is inverted versus live hardware and must not be applied (E-02 / E-03). Second, the OPNsense version is documented as 25.1.12 but was not re-verified live; roughly a dozen references still say 25.7, which is the future CARP target, not the running version (E-23). Third, Randy reports PVE 9.2.3 live while the documentation still says 9.1.1, a drift the prior audit did not catch (E-04, DR-10).

## 4 Infrastructure Inventory

Every entry below is confirmed from live read-only SSH evidence collected 2026-07-23 (E-01 through E-08) unless explicitly flagged as documented-only. Where the live reading disagrees with a repository document, the live value is treated as canonical and the discrepancy is carried into the drift register (Section 18).

### 4.1 Compute and storage hosts

The estate is a seven-node Proxmox VE cluster (km-cluster) plus one standalone Proxmox host (pve1). The two Dell R730 servers and the SuperMicro storage node carry the production research, inference, and storage workloads; the four HP EliteDesk nodes carry the network edge and core services.

| Host | Role                                                      | CPU / RAM       | GPU  | PVE   | Kernel / status                                             |
|------|-----------------------------------------------------------|-----------------|------|-------|-------------------------------------------------------------|
| pve1 | Standalone Pi-hole / homepage host (NOT a cluster member) | Mac Mini, 16 GB | none | 9.1.9 | 7.0.0-3-pve; 142 pending updates; openipmi failed (no BMC)  |
| pve2 | OPNsense host + cluster node                              | i7-8700, 32 GB  | none | 9.2.3 | 7.0.12-1-pve; reboot-required; excluded from hardening role |
| pve3 | Core services + rke2-cp1                                  | i7-8700, 48 GB  | none | 9.2.3 | 7.0.12-1-pve; e1000e TSO/GSO-off mitigation persisted       |
| pve4 | Grafana stack + rke2-cp2                                  | i5-7500T, 32 GB | none | 9.2.3 | 7.0.12-1-pve; sustained load ~5                             |
| pve5 | Headscale, Pi-hole2, rke2-cp3, HAOS                       | i5-7500T, 32 GB | none | 9.2.3 | 7.0.12-1-pve                                                |

| Host      | Role                                           | CPU / RAM                           | GPU               | PVE       | Kernel / status                                     |
|-----------|------------------------------------------------|-------------------------------------|-------------------|-----------|-----------------------------------------------------|
| QuarkyLab | ML / DUNE research / student GPU               | 2× E5-2699 v4 (44c/88t), 512 GB     | 1× RTX 8000 48 GB | 9.2.3     | 6.14.11-9-pve (pinned); kernel + full nvidia-* held |
| Jarvis    | LLM inference                                  | 2× E5-2687W v4, 384 GB              | 2× RTX 6000 24 GB | 9.2.3     | 6.14.11-9-pve (pinned); kernel held only (QL-R12)   |
| Randy     | Storage / PBS / Jellyfin / rke2 storage worker | 2× E5-2690 v3 (24c/48t), 128 GB ECC | RX (transcode)    | 580 9.2.3 | 7.0.12-1-pve; docs say 9.1.1 (DR-10)                |

#### Live confirmations of contested facts

- **GPU assignment (DR-01):** nvidia-smi confirms QuarkyLab carries the single RTX 8000 (46080 MiB) and Jarvis carries two RTX 6000 (24576 MiB each), driver 550.163.01. The review specification's "correction" inverting these is wrong and must not be applied.
- **Kernel pins intact:** both GPU nodes run the pinned 6.14.11-9-pve; the pin protects the NVIDIA 550 stack. Never upgrade the kernel or change GRUB\_DEFAULT on either node (QL-REC20).
- **Randy PVE version (DR-10):** live pveversion reports pve-manager/9.2.3, whereas CLAUDE.md, README, and the fact ledger still state 9.1.1. New drift found by this review.
- **Patch lag:** pending-update counts range 63 (Randy) to 142 (pve1); all cluster nodes report reboot-required (QL-R13).

## 4.2 Network devices

| Device          | Model              | Mgmt IP              | Role / status                                                                                                                                                                                               |
|-----------------|--------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EX3400-48P      | Juniper EX3400     | 192.168.10.50        | Core switch KM-EX3400, JunOS 23.4R2-S7.4, dual PSU both OK, CPU 37C, no alarms, VC-mode enabled (1 member). RSTP root as of 2026-07-23 after bridge-priority 4096 was applied (QL-R23). xe-0/2/3 UP at 10G. |
| USW-24-250W UDR | UniFi UniFi Router | via UDR 192.168.10.2 | Access switch, trunk to ge-0/0/46. VLAN 1 wireless controller / AP - NOT the WAN edge.                                                                                                                      |
| AP7901          | APC                | -                    | Managed PDU on ge-0/0/38.                                                                                                                                                                                   |

The EX3400 was inspected live later in the review, once the operator supplied password credentials (E-22, E-25). Key-based SSH is still not configured, so the automation gap (QL-R16) stands, but switch state above is live-verified rather than inferred. Note also that the previously listed EX2300 lab switch **does not exist and never has** (operator confirmed); it has been removed from the inventory and ge-0/0/45 is simply an unused down port (DR-15).

## 4.3 Power devices

Two UPS units protect the estate (live readings in Section 13): Middle Atlantic UPS-OL2200R (UPS-B, servers and shelf) and Tripp Lite SMART1500VA (UPS-A, switch, UniFi, EliteDesk nodes). Both are online at full charge.

## 4.4 Hardware roadmap

Two hardware changes are in flight and should be tracked as capacity/redundancy items: Randy RAM expansion 128 GB → 192 GB (+64 GB on hand), and a Dell R730xd being commissioned as a second PBS/backup target and Jellyfin transcode node. The R730xd directly addresses the single-backup-node gap (QL-R03, QL-REC03) and its power impact is estimated in Section 13.

## 5 Network Architecture and Correctness

This section reconstructs the network from repository evidence and live cross-checks rather than trusting any single diagram. The EX3400 core switch was inspected live during this review after the operator supplied password credentials (E-22, E-25); that inspection produced one significant new finding and corrected two documented facts. One constraint remains: OPNsense was deliberately not touched because no read-only API credential was used (E-23). Items that rest only on repository documentation, not on a live reading, are marked *inferred*. See diagrams/logical-topology.mmd and diagrams/physical-topology.mmd.

**Live switch inspection, 2026-07-23.** Hostname KM-EX3400, model ex3400-48p, Junos 23.4R2-S7.4. Virtual-Chassis mode enabled with a single member (FPC0 master). Both power supplies OK, CPU sensor 37 C, no active alarms. SNMP is read-only and client-locked to 192.168.10.183/32 with 0.0.0.0/0 restrict. bpdu-block-on-edge is set, with edge on ge-0/0/22, ge-0/0/36 and ge-0/0/41.

## 5.1 VLAN and Addressing Design

The estate runs seven VLANs on the EX3400 (ELS), gatewayed by OPNsense. The design is a clean default-deny segmentation model with a dedicated out-of-band plane.

| ID | Name         | Subnet          | Purpose / population                                   |
|----|--------------|-----------------|--------------------------------------------------------|
| 1  | mgmt/default | 192.168.10.0/24 | Proxmox hosts, corosync, services, switch mgmt (dense) |
| 20 | trusted/OOB  | 192.168.20.0/24 | iDRAC x2 + IPMI (BMCs) + Ares jump leg                 |
| 30 | servers      | 192.168.30.0/24 | NFS/PBS/GPU egress; dual-homed GPU and storage nodes   |
| 40 | IoT          | 192.168.40.0/24 | smart-home, sensors (sparse)                           |
| 50 | VoIP         | 192.168.50.0/24 | CP-8841 phones, FreePBX (deferred)                     |
| 60 | guest        | 192.168.60.0/24 | isolated guest WiFi                                    |
| 70 | lab          | 192.168.70.0/24 | CCNA/experimental; standalone sandbox node             |

The GPU and storage nodes (QuarkyLab, Jarvis, Randy) are dual-homed: management and corosync stay on VLAN 1 while NFS, PBS, and internet egress ride VLAN 30. Corosync was deliberately kept off VLAN 30 to hold the cluster ring on stable L2, which is sound practice.

## 5.2 EX3400 Port Map (live-verified, E-22 / E-25)

| Port                | Attachment                     | Mode          | VLANs                                          |
|---------------------|--------------------------------|---------------|------------------------------------------------|
| ge-0/0/22           | Jarvis nic2 (onboard 1G)       | Trunk         | native 1 (+ tagged 30, now vestigial)          |
| ge-0/0/24           | QuarkyLab nic2                 | Trunk         | native 1 + tagged 30                           |
| ge-0/0/30, /32, /44 | iDRAC/IPMI OOB                 | Access        | 20 only                                        |
| ge-0/0/38           | APC AP7901 PDU                 | Access        | 1                                              |
| ge-0/0/41           | Ares mgmt/jump                 | Dual          | native 1 + tagged 20                           |
| ge-0/0/45           | unused (no neighbor)           | Down          | none                                           |
| ge-0/0/46           | UniFi USW-24 Port 24           | Trunk         | native 1 + tagged 20/30/40/50/60/70            |
| xe-0/2/0            | Randy nic3 (10G ConnectX-3)    | Trunk         | 1 + 30 (storage/NFS)                           |
| xe-0/2/2            | Jarvis enp132s0 (10G ConnectX) | Access        | 30 (dedicated storage/egress)                  |
| xe-0/2/3            | UniFi SFP2                     | Up 10G (DESG) | native 1; link confirmed live (DR-14 resolved) |

Two documented facts were corrected by the live inspection. First, ge-0/0/45 does not connect to an EX2300: the operator confirms **no such switch exists or ever has**, and the port is simply down with no LLDP neighbor. The EX2300 was a documentation phantom and has been removed from the inventory (DR-15). Second, xe-0/2/3 is **up at 10 Gbps** (media type copper, LLDP neighbor a Ubiquiti device) and forwards as a designated port, so the previously recorded EEPROM speed-mismatch fault is stale and the redundant UniFi path does exist (DR-14 resolved).

The ELS native-vlan-id question is now settled by live configuration rather than by narrative: the switch carries set interfaces ge-0/0/22 native-vlan-id 1 and the same on ge-0/0/24, ge-0/0/41, ge-0/0/46, xe-0/2/0 and xe-0/2/3. It is set at the physical-interface level, not under unit 0 family ethernet-switching. A stale note in two documents still claims the EX3400 does not support it at all; that is false and would cause an operator to build a broken trunk (drift DR-08, doc fix D-08).

### 5.3 Gateway, DHCP, and DNS

OPNsense (VM 100 on pve2) is the single LAN router, firewall, DHCP server, and inter-VLAN gateway, live-confirmed running with high egress counters (E-06). DHCP moved to the Kea backend after an empty-Kea outage on 2026-07-20; Kea holds 12 reservations and serves all seven VLAN scopes. WAN is dual-path with single NAT: WAN1 is a Spectrum public block terminated directly on vtnet0 (avoiding double-NAT), WAN2 is a FirstNet 5G hotspot on vtnet2 as an automatic failover gateway group.

DNS is genuinely highly available: primary Pi-hole .177 (pve1 CT103) and secondary .178 (pve5 CT108, a nebula-sync full mirror on a 15-minute timer). OPNsense DHCP hands out both resolvers on every scope, so clients fail over automatically. The resolver chain is client to Pi-hole to Unbound on .1 to internet. A residual concern is that both Pi-holes forward to a single Unbound instance (enterprise finding R-03), which has historically been the root of intermittent DNS behavior. Critical containers (.177 Pi-hole, .181 NPM, .183 Grafana, .148 Homepage) were made static-in-container so a DHCP-server outage cannot drop routing, a good hardening move.

### 5.4 Firewall Policy

The inter-VLAN posture is default-deny with explicit allows, and several clamps have been applied and verified:

- **BMC isolation and egress clamp (verified 2026-07-17):** all three BMCs sit on VLAN 20; only Ares reaches them; a floating quick-block prevents VLAN 20 to internet, so compromised BMC firmware cannot reach VLAN 1, VLAN 30, or the internet.
- **VLAN 30 to VLAN 1 block:** one surgical rule blocks servers to 192.168.10.0/24, with a narrow allow for Prometheus scrape on :9100.
- **DNS pinholes:** isolated VLANs (IoT/VoIP/Guest/Lab) keep :53 to the Pi-hole alias while remaining isolated.
- **Deny logging:** all eight VLAN deny rules log to Wazuh/Loki for lateral-movement visibility.

**Standing decision respected - VLAN 1 egress lockdown is NOT enforced.** The management-plane to internet limit is documented as deliberately deferred, and this review does not recommend enforcing it without first satisfying every prerequisite: UniFi discovery and required multicast/broadcast behavior, legitimate network-device outbound connectivity, RKE2 and container-registry/CDN dependencies, Mist or other vendor callbacks, administrative access paths, DHCP reservations, an emergency access path, current firewall API permissions, and a tested rollback. Enforcing it prematurely is a credible way to cause a self-inflicted outage.

### 5.5 Remote Access and Reverse Proxy

Headscale (self-hosted, LXC 105 on pve5, v0.29.1) is the internal admin mesh only, with nine nodes; students and researchers do not use it. Student and researcher ingress is a Cloudflare Tunnel (quarkylab.kylemason.org) with no inbound ports and the server IP hidden. Public web services are fronted by Nginx Proxy Manager (LXC 101, admin :81 firewalled to Ares only, F-05):

| Hostname               | Backend                    | TLS / auth               |
|------------------------|----------------------------|--------------------------|
| vault.kylemason.org    | Vaultwarden .30.182        | CF DNS-01 / login        |
| grafana.kylemason.org  | Grafana .183 (pve4)        | CF DNS-01 / Grafana      |
| homepage.kylemason.org | Homepage .148              | CF DNS-01 / basic auth   |
| wazuh.kylemason.org    | Wazuh .184                 | CF DNS-01 / Wazuh        |
| chat.netframe.local    | Open WebUI .30.185         | internal / admin acct    |
| llm.netframe.local     | Jarvis llm_router .31:8000 | HTTP, Pi-hole-scoped DNS |

### 5.6 Spanning Tree: Root Was Elected by Accident (Now Remediated)

The most substantive finding from the live inspection concerns spanning tree. The documentation asserted that the EX3400 was the RSTP root at bridge-priority 4096. That was **false**. The committed configuration contained **no bridge-priority statement at all**, leaving the core switch at the default 32768. The UniFi USW-24-250W (b4:fb:e4:1b:ca:3c) had therefore won the root election on the lower MAC address, and the core reached it through ge-0/0/46 at a root cost of 20000 over a 1 Gb link.

This was not an outage, but it is a genuine L2 hardening gap: root placement was decided by MAC accident rather than by design, so a reboot or topology change could relocate the root unpredictably, and traffic paths were computed toward an access switch rather than the central, dual-PSU core.

**Remediated during this review** (operator applied, reviewer verified, E-28). set protocols rstp bridge-priority 4096 was committed. The switch is now genuinely root: Root ID and Bridge ID are both 4096.c0:42:d0:f9:ec:71, and ge-0/0/46 changed role from ROOT to DESG. Worth recording: the first attempt auto-reverted because a commit confirmed was never confirmed. The safety net behaved exactly as intended and the change only persisted once deliberately committed. Tracked as QL-R23 / QL-REC21.

## 5.7 Findings

- **HIGH** OPNsense is a single point of failure for all routing, DHCP, and inter-VLAN traffic (QL-R04). A CARP HA plan exists (QL-REC10).
- **MEDIUM** No automated network-configuration capture (F-N1, R-08, QL-R16). The switch config was captured once by hand during this review; key-based SSH is still not configured and nightly capture is still absent. Deploy Oxidized (QL-REC09).
- **MEDIUM** Single Unbound recursion point on .1 (enterprise R-03).
- **LOW** A residual Amazon Echo/Fire tablet remains on flat VLAN 1 pending the IoT cutover.
- **INFO** IPv6 is not configured on the VLANs and is default-denied; no explicit jumbo/MTU tuning is documented on the 10G fabric.
- **INFO** **Resolved this review:** STP root pinned to the core (QL-R23); xe-0/2/3 confirmed up at 10G (DR-14); EX2300 removed as a phantom asset (DR-15); native-vlan-id placement verified in live config (D-08).

## 6 Proxmox and Virtualization Review

### 6.1 Cluster health

The cluster is healthy and fully quorate. Live pvecm status from pve3 (E-01) reports seven nodes, seven votes, Quorate: Yes, expected votes 7, quorum 4, config version 13, ring ID 1.9eda7. corosync-cfgtool -s shows every peer connected on the single knet link over VLAN 1. Node IDs map to pve2 (.204), pve5 (.203), pve4 (.202), pve3 (.201), QuarkyLab (.179), Jarvis (.31), and Randy (.187). pve1 (.193) is correctly absent - it is a standalone host, not a cluster member.

### 6.2 Node consistency

All seven cluster members run PVE 9.2.3. The four EliteDesk nodes and Randy run kernel 7.0.12-1-pve; the two GPU nodes run the pinned 6.14.11-9-pve. The standalone pve1 lags at PVE 9.1.9 / kernel 7.0.0-3-pve. The only version anomaly inside the cluster is Randy's package set reporting 9.2.3 while documentation still records 9.1.1 (DR-10); functionally the node is healthy.

### 6.3 Guest inventory

Cluster-wide guest state from pvesh get /cluster/resources (E-06):

| ID    | Name             | Node      | Purpose                     | onboot | Backup        |
|-------|------------------|-----------|-----------------------------|--------|---------------|
| CT101 | nginx-proxy      | pve3      | Reverse proxy (NPM)         | 1      | randy-pbs     |
| CT102 | vaultwarden      | pve3      | Secrets manager             | 1      | randy-pbs     |
| CT103 | grafana          | pve4      | Metrics/logs/alerting stack | 1      | randy-pbs     |
| CT105 | headscale        | pve5      | Tailnet coordinator         | 1      | randy-pbs     |
| CT106 | homepage         | pve3      | Dashboard                   | 1      | randy-pbs     |
| CT107 | openwebui        | pve3      | Chat UI to llm_router       | 1      | randy-pbs     |
| CT108 | netframe-pihole2 | pve5      | Secondary DNS               | 1      | randy-pbs     |
| VM100 | OPNsense         | pve2      | Router / firewall / DHCP    | 1      | randy-pbs     |
| VM104 | wazuh            | QuarkyLab | SIEM                        | 1      | randy-pbs-10g |
| VM110 | homeassistant    | pve5      | Home automation (HAOS)      | 1      | randy-pbs     |
| VM201 | rke2-cp1         | pve3      | Kubernetes control plane    | 1      | randy-pbs     |
| VM202 | rke2-cp2         | pve4      | Kubernetes control plane    | 1      | randy-pbs     |
| VM203 | rke2-cp3         | pve5      | Kubernetes control plane    | 1      | randy-pbs     |

Every production guest has `onboot=1` and a nightly backup job. The cluster-side counterpart to CT103 Grafana and CT105 Headscale confirms both moved to pve4 and pve5 respectively (the 2026-07-16 de-concentration), contradicting README/topology which still draw them on pve3 (DR-03, DR-04).

## 6.4 High availability, replication, and storage definitions

`ha-manager` status reports quorum OK but fencing standby (CRM watchdog standby) with no HA groups defined (E-06). A node failure therefore does not auto-migrate guests; recovery is manual (QL-R18) - a defensible choice on consumer EliteDesk hardware, but it should be an explicit, documented decision. `pvesr` status shows no replication jobs.

Two PBS storage targets are defined in `/etc/pve/storage.cfg`: `randy-pbs` (server 192.168.10.187, `keep-all=1`) carries the LXCs and VM100/VM110/rke2; `randy-pbs-10g` (server 192.168.30.187 over the 10G VLAN 30 path, `keep-daily=7`, `keep-weekly=4`, restricted to nodes QuarkyLab/Jarvis/Randy) carries the Wazuh VM. Both point at the same datastore and fingerprint. Enterprise repositories are disabled and the no-subscription repo is enabled on cluster nodes.

## 6.5 Orphans, stale objects, and discipline

The only orphan of note is CT104 on pve1 (old homepage), which documentation records as retired 2026-06-24 but which is still running (DR-11, R-12). No orphaned disks, stale snapshots, or abandoned templates were surfaced by the live captures.

### Strength: kernel-pin discipline

The GPU nodes hold their kernel via GRUB\_DEFAULT and (on QuarkyLab) the full `nvidia-*` package set, deliberately protecting a validated driver stack from an incidental apt upgrade. This is mature change control. The one gap is that Jarvis holds only the kernel, not the `nvidia-*` packages - it should mirror QuarkyLab's fuller hold set (QL-R12, QL-REC12).

# 7 Storage and Data Integrity

## 7.1 Live pool health

Five ZFS pools were inspected live (E-02, E-03, E-04). Four are healthy; one carries a faulted device.

| Pool      | Host      | Size   | Geometry                                      | State                     |
|-----------|-----------|--------|-----------------------------------------------|---------------------------|
| datastore | Randy     | 36.7 T | 4× RAIDZ2 (three 6-wide + one 4-wide Seagate) | ONLINE, 0 errors          |
| bulk      | Randy     | 80.0 T | 3× RAIDZ2 (8+8+6-wide)                        | DEGRADED - mpathv FAULTED |
| workspace | QuarkyLab | 10.9 T | 6-wide RAIDZ1 + 1 hot spare (PERC H330 JBOD)  | ONLINE, 0 errors          |
| tank      | Jarvis    | 9.09 T | 5-wide RAIDZ1 (HBA330 IT)                     | ONLINE, 0 errors          |
| scratch   | Jarvis    | 186 G  | single SSD                                    | ONLINE, 0 errors          |

The live `datastore` layout confirms the disputed documentation fact D-04: the fourth vdev (`raidz2-3`) is four Seagate drives by-WWN, in-pool, with no unallocated spares. The `bulk` geometry confirms D-05: three vdevs totalling 80 T, superseding the stale "2×8-wide / 58.2 T" still in topology and `Proxmox Cluster.md`.

## 7.2 The faulted bulk drive

**MEDIUM QL-R01 - reduced parity, corrected impact.** `zpool status bulk` (E-04) shows device `mpathv` in vdev `raidz2-2` as FAULTED with 3 read / 14 write errors and "too many errors". The device is a SEAGATE ST4000NM0023 (4 TB SAS), WWN 5000c500631a54fb; both SAS paths are dead, so this is a failed physical drive, not a multipath flap. It corresponds to the DS4246 slot-15 drive that failed 2026-07-21 (incident NF-INC-2026-07-22). Follow-up inspection confirmed the drive is now fully offline: it has no `/dev/mapper` entry at all, and ZFS is holding it only by cached label.

Key mitigating facts, all live-verified: the pool reports errors: No known data errors; RAIDZ2 tolerates one drive loss per vdev, so `raidz2-2` still has one parity drive of protection remaining; the last scrub (Sun Jul 19) completed with zero errors; and two DS4246 bays are free for an immediate `zpool replace`.



**Correction: this is loss of redundancy, not imminent data loss**

An earlier draft of this review rated the drive urgent on the basis that bulk held researcher fernanda's DUNE data with no second copy. **Live inspection (E-29) contradicts that**, and the original framing came from stale documentation. `zfs list -r bulk` shows `bulk/fernanda`, `bulk/media`, `bulk/misc` and `bulk/archive` are all 205 to 222 KB placeholders. The only real data is `bulk/backups` at 57.3 GB, which holds the Ares restic repository and a `jarvis-netframe` backup - both themselves secondary copies whose sources still exist. The DUNE research data has **not** been migrated onto bulk. Replacing `mpathv` is therefore prudent maintenance to restore full double parity, not a race against data loss. Two conditions keep it on the list rather than closing it: the exposure becomes real the moment research data migrates onto bulk, so the second copy (QL-R02) must exist *before* that migration; and the Ares restic backup currently targets this degraded pool. Recommended action QL-REC01, reclassified to next maintenance window and pending operator hardware.

### 7.3 Controllers and multipath topology

Randy uses two distinct controllers by design: an AVAGO 3108 MegaRAID for the boot mirror (relocated 2026-07-01 after its original slot died - do not reuse the dead slot) and an LSI 9207-8e in IT mode for the DS4246 data path. The DS4246 is dual-SAS cabled with multipath, so every shelf drive presents as an `mpathX` device. The single enclosure is itself a single point of failure for bulk (H-09, QL-R05 context): loss of the shelf or its cabling takes the whole pool offline.

### 7.4 Redundancy posture and aged media

The two GPU-node pools are RAIDZ1 (QL-R21): workspace tolerates one loss but carries a hot spare; tank tolerates one loss with no spare; scratch is a single SSD with no redundancy (scratch by intent). Several disks are aged SAS. RAIDZ1 on old disks carries resilver-window risk - a second fault during rebuild loses the pool. This is an accepted risk today given clean scrubs and low utilisation, but future rebuilds should prefer RAIDZ2 and keep spares on hand.

### 7.5 Data integrity and capacity

Scrubs are scheduled and recent (datastore and bulk both scrubbed Jul 19, zero repaired). Scrutiny monitors roughly 79 drives across Randy, QuarkyLab, and Jarvis. The alerting question paired with this pool is now settled: the gap was not Scrutiny or delivery but the absence of any device-level ZFS metric, since closed (QL-R11, Section 11). Utilisation is low across the board - datastore 176 G of 36.7 T, bulk 80.7 G of 80 T, workspace 144 G of 10.9 T - so there is no near-term capacity pressure; the risk is redundancy and backup, not space.

### 7.6 Recoverability by dataset

Guest images and configs are recoverable from PBS within roughly 24 h RPO (Section 8). The exceptions are the two datasets with no independent copy: bulk (which today holds only `bulk/backups`, research data having not yet migrated) and `pve1's CT103/CT104`. Ares' own restic backup, notably, targets `sftp:randy:/mnt/bulk/backups/ares` - it lands on the degraded bulk pool, so the workstation backup shares the same at-risk failure domain and should be repointed once a second target exists.

## 8 Backup and Disaster Recovery

Backup execution is healthy and current; backup *architecture* is the weak point. Live evidence (E-08) shows 14 snapshots dated 2026-07-23 from the previous night's runs, with the PBS datastore at 97 GB used of 23 TB. Every cluster guest is protected daily. The problem is that all of those copies live on one node.

### 8.1 Backup Posture

| Asset                  | Target               | Schedule / re-tention    | Notes                                  |
|------------------------|----------------------|--------------------------|----------------------------------------|
| LXC<br>101/102/106/107 | randy-pbs<br>.10.187 | 02:00<br>daily,<br>7d/4w | NPM, Vaultwarden, Homepage, Open WebUI |

| Asset                          | Target                                                  | Schedule / re-<br>tention | Notes                                              |
|--------------------------------|---------------------------------------------------------|---------------------------|----------------------------------------------------|
| LXC 103 Grafana                | randy-pbs                                               | 02:00 daily               | observability stack                                |
| LXC 105 Headscale, 108 pihole2 | randy-pbs                                               | 02:00 daily               | pihole2 mirrors CT103 config                       |
| VM 100 OPNsense                | randy-pbs + config repo                                 | 03:00 daily + 03:17       | config also age-encrypted to GitHub                |
| VM 104 Wazuh                   | randy-pbs-10g .30.187                                   | 03:00 daily               | 10G path                                           |
| VM 110 HomeAssistant           | randy-pbs                                               | 03:30 daily               | HAOS                                               |
| VM 201/202/203 RKE2            | randy-pbs                                               | 04:00 daily               | etcd not separately backed up                      |
| QuarkyLab workspace            | PBS host backup                                         | 05:30 daily               | fernanda refreservation floor                      |
| pve1 CT103/CT104               | <b>HIGH</b> NONE                                        | never                     | primary Pi-hole + old homepage unbacked (QL-R06)   |
| bulk pool data                 | <b>HIGH</b> NONE                                        | never                     | media + fernanda research, no second copy (QL-R02) |
| Ares /home                     | restic to                                               | daily 04:03               | encrypted, but lands on the DEGRADED bulk pool     |
| Ares ~/.claude                 | randy:/mnt/bulk/...<br>randy:/datastore/<br>ares-claude | hourly                    | memory/transcripts, secrets excluded               |

## 8.2 3-2-1 Analysis

The estate is effectively 1-1-1: one PBS instance, on one ZFS pool, in one rack, on one UPS bus (QL-R03, **HIGH**). Three specific exposures compound it:

- **pve1 guests are unprotected** (QL-R06 / R-03): the primary Pi-hole CT103 and the retired CT104 have no vdump job. This is partly mitigated because the secondary Pi-hole CT108 is backed up and nebula-syncs the primary's config, so DNS state is rebuildable.
  - **The bulk pool has no second copy** (QL-R02): it is neither a PBS source nor a target, so the DUNE research data and media it holds exist only on the currently degraded DS4246 shelf.
  - **The Ares restic repository lands on bulk** (randy:/mnt/bulk/ backups/ares), the same degraded pool, so the workstation backup shares the shelf's fault domain.
- Encryption is partial: the OPNsense config backup is age-encrypted and the Ares restic repository is encrypted, but the PBS datastore itself is not encrypted at rest.

### Do not assume a green job is a recoverable backup

PBS reports success and runs a weekly verify plus garbage collection, which is good, but no full restore has been rehearsed for most guests. A backup that has never been restored is an untested backup.

## 8.3 Recovery Priority and Objectives

Recovery priority, highest first: (1) power and core network, (2) OPNsense routing and DHCP, (3) DNS, (4) Randy storage and PBS, (5) Vaultwarden and certificates, (6) the app and AI tiers, (7) research and student services. Estimated objectives from the backup matrix: RPO is 24 h for guest images (daily vdump) and 1 h for Ares memory; RTO ranges from roughly 10-30 minutes per guest restore to several hours for a full cluster rebuild. The OPNsense config RPO is effectively 24 h with a 15-30 minute restore.

## 8.4 Non-Destructive Restore Validation Plan

No restore was performed during this review. The recommended validation, all non-destructive and to be scheduled with operator approval, is: (a) run a PBS verify job and confirm zero failures; (b) restore one representative guest to a scratch VMID on a node with free space, boot it isolated, confirm it starts, then destroy the scratch guest; (c) test-decrypt the OPNsense age config and the break-glass snapshot. Never restore into a production VMID.



## 8.5 Per-Domain Recovery Procedures

The estate has an unusually strong recovery-documentation culture; the procedures below already exist and were reviewed:

- **OPNsense:** serial console via `qm terminal 100` from pve2 plus a DR-tested age-encrypted `config.xml` restore; SSH is disabled by design so console is the only shell.
- **DNS:** `pct start 103` on pve1, or fail to the secondary .178; `pihole restartdns` if the container is up but DNS is stuck.
- **Cluster/secrets loss:** the break-glass age snapshot held on both Ares and Randy breaks the Vaultwarden-in-cluster circular dependency (QL-R08), letting an operator read credentials with Vaultwarden, NPM, and the cluster all down.
- **Network device:** the EX3400 config is not automatically captured, so switch rebuild is slow and manual (gap, QL-REC09).
- **RKE2:** rebuild rests on etcd quorum; etcd is not separately snapshotted beyond VM-level backup.
- **Git and docs:** GitHub plus local clones; the repos are the source of truth for config-as-code.

## 8.6 Recommendations

Add a second, independent backup target (QL-REC03): the planned R730xd running its own PBS with PBS-to-PBS remote sync, plus an offsite copy to Backblaze B2 for the datastore and `bulk/fernanda`. Add the pve1 `vzdump` job (QL-REC04). See `diagrams/recovery-dependencies.mmd`.

# 9 Kubernetes, RKE2, and Containers

## 9.1 RKE2 Control Plane

RKE2 runs a three-node HA control plane on VMs 201/202/203 (pve3/pve4/pve5, .51 - .53), all live-confirmed running (E-06). The API is fronted by a kube-vip VIP at `192.168.10.54:6443`; the CNI is Cilium and load balancing is MetallB over the .71 - .75 pool. etcd runs as an embedded quorum across the three control-plane nodes. The cluster is intentionally CPU-only: GPU scheduling is deferred because SLURM and Ollama own the cards, and running RKE2 GPU scheduling on the same single card would contend for exclusive device access.

Randy joined as a bare-metal storage worker with kubelet reservations sized to measured hardware and a `node.netframe.io/role=storage:NoSchedule` taint, so only tolerating infrastructure DaemonSets land there and ZFS/PBS are not cgroup-throttled. Persistent storage is an `nfs-client` StorageClass backed by Randy datastore/k8s. A documented operational detail worth preserving: agents register on the supervisor port `:9345`, not the apiserver `:6443`.

## 9.2 Container Gaps

- **MEDIUM** No NetworkPolicies: the pod network is flat with no default-deny (QL-R15 / K8S-02).
- **MEDIUM** No Pod Security Admission (K8S-03), and Kubernetes secrets are likely unencrypted at rest (K8S-04).
- **MEDIUM** The private registry is anonymous and a single replica (QL-R14 / K8S-01); its TLS is provided by step-ca, and a `RegistryCertExpiringSoon` alert exists.
- **LOW** Image pinning and floating latest tags were not exhaustively audited; pinning discipline should be confirmed as workloads grow.
- **INFO** etcd is protected only at the VM-backup level; a dedicated etcd snapshot schedule would speed control-plane recovery.

The recommended treatment (QL-REC15) is to add default-deny NetworkPolicies, Pod Security Admission, secrets encryption, and registry authentication before the cluster carries meaningful workloads. None of this should be applied without a maintenance window and rollback.

## 9.3 LXC and Docker Estate

The operator convention is one service per LXC and "no Docker unless explicitly requested," which keeps blast radius small and is a genuine strength. The running guests were enumerated live (E-06): NPM (CT101), Vaultwarden (CT102), Grafana stack (CT103 on pve4), Headscale (CT105), Homepage (CT106), Open WebUI (CT107), secondary Pi-hole (CT108), plus VMs for OPNsense, Wazuh, Home Assistant, and the three RKE2 control-plane nodes. Docker appears only inside a few LXCs where it is warranted: the Grafana stack (Grafana, Prometheus, Loki, InfluxDB, Scrutiny) in CT103, Vaultwarden in CT102, and the Homepage dashboard in CT106.

All guests are set `onboot=1`. Compose-versus-deployed drift could not be fully verified read-only, since that would require exec into each container; it should be spot-checked during normal maintenance. Consistent with the read-only mandate, this review did not pull images, recreate containers, or restart any workload.

The one duplicate to resolve is CT104 on pve1, an old homepage instance that documentation says was retired on 2026-06-24 but that is still running (DR-11 / R-12). Confirm it is the retired skeleton, then stop and disable it (QL-REC17).

## 10 AI and Jarvis Architecture Review

This section audits the netframe-monitor / Jarvis AI-operations system as a production operational system, not as a demo. The central question is whether a language model is ever permitted to state a conclusion as fact, take an action, or become a command without a deterministic gate and explicit human approval. The short answer, verified against the code (E-14), is no: the design is safety-first, and it is one of the signature strengths of this lab.

### 10.1 Architecture and data flow

The system separates a deterministic plane from a model plane, and the deterministic plane is load-bearing for every safety-critical function.

- **Collection (no model).** `netframe_monitor.py` runs every 15 minutes via `netframe-run.sh`, SSHing read-only probes across seven nodes with a dedicated `monitor_key` and writing `last_run.json` plus an append-only `history.jsonl`. It includes end-to-end user-impact probes (LLM\_ROUTER through NPM, and LLM\_ROUTER\_CONFORMANCE reporting config, runtime, and firewall as three independent dimensions that are never collapsed).
- **Report paths.** `netframe_interpret.py` (now), `netframe_daily.py`, `netframe_predict.py` (deterministic least-squares forecast), `netframe_monthly.py`, and `netframe_chief.py` (weekly synthesis). Every path imports both `netframe_policy` and `netframe_evidence` and ends with the same sequence: model prose, then a deterministic evidence section, then `netframe_policy.enforce()`.
- **Admission screen.** `netframe_admission.py` vetoes resource-consuming probes on 72B residency, GPU-busy, interactive user, maintenance window, or minimum interval, and fails closed when GPU state is unknown, so the monitor never becomes the workload it monitors.
- **Deterministic alerter.** `netframe_alert.py` fires node-down Discord messages straight from `last_run.json` on UNREACHABLE transitions, independent of Grafana, using fixed templates (nothing for a model to influence).
- **Config drift.** `netframe_confdrift.py` fingerprints interfaces, ssh, sysctl, and env over the read-only path and diffs against a human-blessed baseline.
- **Interfaces.** `netframe_console.py` serves a read-only HTTP API with no execute endpoint; both the console and report pages are iptables-locked to localhost plus NPM.
- **Model routing.** `llm_router.py` defaults to local Ollama `qwen2.5:72b`, serves a rag model over a numpy cosine index, and escalates to `claude-opus-4-8` on an explicit flag, a `claude- model`, or local failure.
- **Audit.** `netframe_audit.py` maintains a hash-chained, tamper-evident, append-only ledger mirrored to Loki at write time.

### 10.2 AI-safety control matrix

| Control                       | Implementation (evidence E-14)                                                                                                                                                                                                     | Verdict |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| Single shared evidence engine | <code>netframe_evidence.py</code> is called by all six paths; the Findings contract forbids the model stating a confidence; <code>strip_model_confidence</code> removes any residual confidence text on the free-form console path | Strong  |
| Provenance and evidence age   | Every factor carries provenance; <code>freshness()</code> with <code>FRESHNESS_MAX_S = 5400s</code> caps confidence at 40 and renders STALE; unverifiable citations contribute zero and trigger a 30 ceiling                       | Strong  |
| Deterministic confidence      | <code>evidence_quality</code> and confidence are pure-stdlib floors-plus-ceilings over telemetry; the module states "no model is consulted here"; tightest ceiling wins                                                            | Strong  |
| Human-gated remediation       | <code>netframe_remediate.py</code> : 3-item Tier-1 allowlist only, list-form argv (no shell), requires explicit approve plus confidence $\geq 95$ ; anything else refuses and prints a manual workflow                             | Strong  |
| Universal policy screening    | <code>netframe_policy.enforce()</code> is the single gate on every model-to-operator path, including chat; the Discord alerter is deterministic templates with nothing to bypass                                                   | Strong  |
| Model output never a command  | <code>_guard_actions</code> strips shell commands and code fences from chat output and honors only an allowlisted <code>[action:id]</code> tag; the console, not the model, renders the gated command                              | Strong  |
| Untrusted telemetry           | Telemetry wrapped in BEGIN/END UNTRUSTED markers with a data-not-instructions prompt, plus a deterministic injection tripwire (honest caveat: a tripwire for overt imperatives, not a boundary)                                    | Good    |

| Control                     | Implementation (evidence E-14)                                                                                                                             | Verdict |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| Model-offline survivability | Collector, alerter, confdrift, admission, forecast tables, and evidence scoring are all no-model; node-down alerts fire before any LLM step                | Strong  |
| Failed model calls          | Every path degrades to a deterministic fallback; <code>policy.enforce</code> fails LOUD but not closed (emits an UNSCREENED stamp rather than suppressing) | Strong  |

### 10.3 Maturity by function

| Function             | Rating                | Basis                                                                                                      |
|----------------------|-----------------------|------------------------------------------------------------------------------------------------------------|
| Monitoring assistant | Strong / production   | Deterministic 15-min collector, dead-man heartbeat, Grafana-independent alerter, locked read-only surfaces |
| Incident analysis    | Good                  | Deterministic blast-radius from a knowledge graph, two-axis evidence and confidence per finding            |
| Capacity planning    | Solid (deterministic) | Least-squares slope and days-to-90%; math never delegated to the model                                     |
| Documentation        | Strong                | Auto now/daily/weekly/monthly reports, hash-chained ledger, unified memory dashboard                       |
| Change planning      | Recommend-only        | Chief P0-P3 ranking; propose emits a manual workflow; confdrift human-bless gate                           |
| Remediation          | Minimal / human-gated | 3-action allowlist, confidence gate, never autonomous (constitution <code>authority_limits.md</code> )     |

### 10.4 Code versus documentation

The `jarvis-ai-ops` LaTeX corpus is explicitly propose-and-assess. The code is now ahead of those assessments on the two biggest historical gaps: the two-axis evidence scoring proposed in NF-AIOPS-005 is fully implemented, and the tamper-evident audit ledger flagged as open in JAR-12 is built. Two residual gaps remain and are reflected in the recommendations:

- **Remediation approval gate (JAR-08, QL-REC13).** Report and chat confidence are now deterministic, but the Tier-1 execution gate still reads a caller-supplied confidence integer rather than the `netframe_evidence` engine. The gate is safe by construction (allowlist plus explicit approval) but the number is not yet wired to computed evidence.
- **Config conformance.** `confdrift` compares production to a blessed snapshot of production, not to Git. The specific `llm_router` env-drift incident class is now covered by the separate conformance probe, but a general conform-to-Git target is not yet built.

#### Assessment

As a home-lab AI-operations platform this is unusually mature. The trust foundation - single deterministic evidence engine, universal policy screen, human-gated allowlist remediation, untrusted-input handling, and full model-offline survivability - is enterprise-grade. Critical alerting does not depend on any single AI model, and no model output can become an action or a shell command without a deterministic gate and explicit approval. The remaining lift is the intelligence layer (evaluation harness, retrieval calibration, Git conformance), not the safety layer. This directly satisfies the review requirement that deterministic monitoring remain available if the AI is offline and that remediation cannot occur without approval.

### 10.5 Credentials the AI touches

Reported by type and location only, values redacted (E-14). `ANTHROPIC_API_KEY` in `/etc/llm_router.env` (0600, unset means local-only); `DISCORD_BOT_TOKEN` and the operator user id in `/etc/netframe-alert.env` (root 0600); the read-only `monitor_key` used by the collector, confdrift, and the scoped wazuh-restart wrapper; and the restic backup passphrase file. The restic passphrase is the one single-point risk: if it lives only on Jarvis, loss of that host makes the encrypted backup unreadable, so it must also live in Vaultwarden. No hardcoded secrets were found in any source file read.

## 11 Monitoring, Logging, and Alerting

### 11.1 Stack and topology

Observability is a Grafana, Prometheus, Loki, InfluxDB, and Scrutiny stack running together in CT103 on pve4 (192.168.10.183), live-confirmed (E-06). Alerting uses Grafana Unified Alerting straight to Discord with no Alertmanager.

Two contact points exist: an infrastructure channel (discord-alerts) and a UPS channel (discord-ups). Prometheus was reported at 15 of 15 targets up (E-15), which means the silent not-scraping failure mode is clean. Config lives as code in a private repository, which is a genuine operational-maturity strength.

## 11.2 Alert inventory

The following rules were extracted from the provisioning repository (E-15). GAP- rows are missing alerts, not deployed ones.

| Alert                     | Source               | Sev  | Threshold                               | Disposition                    |
|---------------------------|----------------------|------|-----------------------------------------|--------------------------------|
| ClusterNotQuorate         | corosync textfile    | crit | quorate < 1, 2m                         | Keep                           |
| ClusterVotesBelowExpected | corosync textfile    | warn | votes deficit > 0, 5m                   | Keep                           |
| InstanceDown              | node/blackbox        | crit | up < 1, 2m                              | Keep                           |
| PiholePrimaryDown         | blackbox DNS         | crit | probe .177 < 1, 2m                      | Keep                           |
| PiholeSecondaryDown       | blackbox DNS         | warn | probe .178 < 1, 2m                      | Keep                           |
| ZfsPoolDegraded           | zpool textfile       | crit | health > 0, 5m                          | Keep; blind to device faults   |
| ZfsDeviceFaulted          | zpool textfile       | crit | non-ONLINE devices > 0, 5m              | <b>Added 2026-07-23</b>        |
| GpuTempHigh               | nvidia-smi textfile  | warn | temp > 87C, 5m                          | Keep                           |
| GpuMemoryHigh             | nvidia-smi textfile  | warn | VRAM used > 90%, 10m                    | Keep                           |
| DiskAlmostFull            | node_filesystem      | warn | root > 90%, 10m                         | Extend to pools                |
| LowMemory                 | node_memory          | warn | avail < 8%, 10m                         | Keep                           |
| RegistryCertExpiringSoon  | blackbox tls         | warn | cert < 12h, 15m                         | Broaden to all certs           |
| BackupVerifyReportStale   | Randy textfile       | warn | age > 26h, 15m                          | Keep (caught mpathv)           |
| BackupVerifyFailing       | Randy textfile       | warn | pass < 1, 5m                            | Consider critical              |
| AresBackupStale           | Randy textfile       | warn | age > 26h, 15m                          | Keep                           |
| NetframeMonitorStale      | Jarvis heartbeat     | warn | age > 1h, noData alerting               | Keep (dead-man)                |
| UPS battery low           | PeaNUT               | crit | charge < 50, 2m                         | Keep                           |
| UPS runtime low           | PeaNUT               | crit | runtime < 300s, 1m                      | Keep                           |
| PrimaryWANDown            | pve2 textfile        | crit | gateway < 1, 1m                         | Keep                           |
| FirewallBlockRateSpike    | Loki filterlog       | warn | > 500 / 5m                              | Keep                           |
| NetworkDeviceConfigChange | Loki syslog          | warn | commit > 0                              | Keep                           |
| NetworkSyslogSilent       | Loki stream          | warn | < 50 / 15m, noData alerting             | Keep (dead-man)                |
| GAP-quorum-guest          | (needs pve-exporter) | -    | per-guest down                          | Missing (QL-REC06/07)          |
| GAP-pbs-verify-stale      | -                    | -    | PBS verify failed/stale                 | Missing (QL-REC07)             |
| GAP-host-auth-failure     | -                    | -    | repeated auth failures                  | Missing (DETECT-01c)           |
| GAP-cert-expiry-broad     | -                    | -    | any internal/LE cert near expiry        | Missing (only registry to-day) |
| GAP-2nd-channel           | -                    | -    | alert egress independent of WAN/Discord | Missing (QL-REC14)             |

## 11.3 Coverage

| Signal         | Status  | Notes                                                                                           |
|----------------|---------|-------------------------------------------------------------------------------------------------|
| Availability   | Covered | InstanceDown, Pi-hole probes, PrimaryWANDown, quorum                                            |
| Latency        | Partial | OPNsense per-WAN latency on dashboard only; DNS probe is up/down                                |
| Errors         | Partial | Journal errors as a dashboard chip; FirewallBlockRateSpike; no app error-rate rule              |
| Saturation     | Covered | LowMemory, GpuMemoryHigh; no CPU-saturation rule                                                |
| Capacity       | Partial | DiskAlmostFull on root only; pool/non-root capacity not alerted                                 |
| Temperature    | Covered | GpuTempHigh; no CPU or ambient temp alert                                                       |
| Power          | Covered | UPS battery and runtime via PeaNUT                                                              |
| Storage health | Covered | ZfsPoolDegraded plus new ZfsDeviceFaulted (device-level, added this review) plus Scrutiny SMART |
| Cert expiry    | Narrow  | Registry TLS only; other certs not monitored                                                    |
| Backup success | Covered | BackupVerify stale/failing, AresBackupStale                                                     |
| Auth failures  | Gap     | No host auth-failure alert rule (dashboard chip only)                                           |
| Config drift   | Partial | Hardening-drift chip and NetworkDeviceConfigChange; no host config-drift alert                  |

## 11.4 Blind spots and self-dependency

- **Single alert egress (QL-R09, DETECT-01d).** All alerts leave over one Discord webhook across the WAN. A WAN or Discord outage blinds alerting, and the deterministic node-down message shares that path. A second out-of-band channel is recommended (QL-REC14).

- **Co-located stack (QL-R10).** Metrics, logs, and alert evaluation all run in one CT on one node. Losing that node loses monitoring and stops alert evaluation.
- **Loki unauthenticated.** `auth_enabled: false` on VLAN 1 makes logs readable and poisonable by any VLAN 1 host.
- **SIEM scope (QL-R22).** Wazuh agents run only on the eight hypervisors; guests, OPNsense, the switch, and endpoints are blind to the SIEM.

### The silent-alert lesson, root-caused and closed

The degraded bulk pool was masked for roughly 71 hours, and the condition was ultimately surfaced by `BackupVerifyReportStale` rather than by `ZfsPoolDegraded`. This review determined the cause, and it was **not** an alert-delivery failure.

The collector `zfs-pool-textfile.sh` emitted only `node_zfs_pool_health`, derived from `zpool list -o health`. That property returns `ONLINE` for bulk because the `raidz2` vdev still holds redundancy, so the metric read 0 (healthy) the entire time. The proof is unambiguous: `max_over_time(node_zfs_pool_health{pool="bulk"}[7d])` evaluated to 0, so the rule, which fires on `> 0`, could never have triggered. No per-device metric existed anywhere in the stack. A faulted disk inside a still-redundant pool was simply invisible to alerting (QL-R11).

**Remediated 2026-07-23.** The collector now also emits `node_zfs_pool_nonhealthy_devices{pool}`, a count of vdev devices not `ONLINE`, and a new critical rule `ZfsDeviceFaulted` fires on `> 0` for 5m to the infra channel. It is deployed to Randy, QuarkyLab and Jarvis and verified live in Prometheus: `bulk = 1`, every other pool = 0. This rule would have paged on day one. Config-as-code lives in `netframe-monitoring-stack` branch `fix/zfs-device-level-alert`.

A second problem surfaced while deploying it: the collector running on all three nodes had drifted ahead of the repository, carrying an uncommitted `node_zfs_pool_capacity_percent` metric. Deploying the repository copy unchanged would have silently deleted a live metric. It was reconciled into the repository first (DR-17). The backup-verify scheduler itself has separately been moved to a persistent systemd user timer.

## 11.5 What is done well

Dead-man heartbeats already exist (`NetframeMonitorStale`, `NetworkSyslogSilent`, both alerting on `noData`), a `corosync` quorum textfile alert has been added, WAN-down detection is wired from OPNsense `dpinger`, and the entire configuration is version-controlled. The recommended next steps are a read-only `prometheus-pve-exporter` for guest and quorum metrics (QL-REC06), the backup-stale, verify-failed, and broad cert-expiry rules (QL-REC07), and the second alert channel (QL-REC14).

## 12 Security Architecture Review

This is a defensive review only. No credential was used, no exploit was attempted, and all secret values are redacted to type, path, and a redacted indicator.

### 12.1 Network segmentation (strong)

Segmentation is the standout security control. Seven VLANs are defined and live on the EX3400 (E-21, architecture digest). Both server-grade BMCs and the Randy IPMI were moved off flat VLAN 1 to VLAN 20 with factory credentials rotated, and an egress clamp (applied and verified 2026-07-17) prevents a compromised BMC from reaching VLAN 1, VLAN 30, or the internet. A single surgical rule blocks VLAN 30 to VLAN 1 except a narrow Prometheus scrape. IoT, VoIP, Guest, and Lab are isolated with a DNS pinhole so they keep ad-blocking without lateral reach. All eight deny rules log to Loki and Wazuh. There is no inbound port exposure; external reach is via NPM with TLS and via a Cloudflare Tunnel with no open ports. The VLAN 1 egress lockdown is deliberately deferred as disproportionate gold-plating, which this review respects and does not recommend enforcing without the documented prerequisites.

### 12.2 Identity and access

Administration uses shared root over key-only SSH. The Ansible hardening role installs a strong `sshd` policy (key-only, `PermitRootLogin prohibit-password`, no password or keyboard-interactive) but has been run in check mode only and is not yet enforced on the nodes (DR-12), so the fleet is not yet hardened in production. There is no MFA on administrative interfaces, and the prior assessment flagged weak Ares admin credentials (enterprise R-23). These are reasonable for a single-operator lab but are the items an enterprise reviewer would raise first.

## 12.3 Secrets

The committed-secret posture is clean: the Home-Lab full-history gitleaks scan is clean across 300-plus commits, and every tracked hit is a placeholder or shell-variable reference (E-17). This review adds three findings the committed-secret audit could not see, all at rest on the Ares workstation and none ever committed to git:

| Type                 | Path                  | Redacted | Status                                                      |
|----------------------|-----------------------|----------|-------------------------------------------------------------|
| GitHub PAT (classic) | ~/github_helper.py:4  | ghp_...  | Untracked, gitignored, never in history; live token at rest |
| Generic API key      | ~/apikey.txt          | key=...  | Untracked; real value on disk                               |
| Wazuh API password   | ~/wazuh-api-newpw.txt | (0600)   | Untracked; plaintext at rest                                |

The GitHub PAT (QL-R07) should be rotated and moved to an environment or credential store, and the two files removed once their values are relocated (QL-REC05). Separately, six repositories lack a .gitignore and the tracked .bash\_aliases and .ssh/config publish internal topology in a public-intent superproject (QL-REC18).

## 12.4 Unauthenticated services and Kubernetes posture

Ollama and llm\_router accept unauthenticated LAN requests but are iptables-locked to known clients; the private registry is anonymous (K8S-01) and Loki is unauthenticated (QL-R14). RKE2 runs with a flat pod network, no NetworkPolicies, no Pod Security Admission, and likely unencrypted secrets (QL-R15, K8S-02/03/04). These are lateral-movement value if a LAN host is compromised rather than direct external exposure. TLS is a strength: grade-A internal certificates from step-ca and Let's Encrypt via Cloudflare DNS-01. Patch lag is tracked separately (QL-R13).

## 12.5 Reconciliation with the prior security assessment

| ID             | Sev  | Finding                                   | Status                                   |
|----------------|------|-------------------------------------------|------------------------------------------|
| F-001          | High | Untrusted IoT/printer/mobile on mgmt VLAN | Open (needs UDR admin)                   |
| F-003          | High | Workstation cleartext FTP / EOL SMB/NFS   | Open                                     |
| F-006          | High | Unencrypted cluster-root SSH keys         | Open                                     |
| K8S-01         | High | Anonymous private registry                | Open                                     |
| NET-01         | Med  | OOB/BMC asymmetric isolation              | Done (VLAN 20 egress clamp)              |
| VIRT-01        | Med  | Hypervisor firewall disabled              | Partial                                  |
| VIRT-02        | Med  | Mgmt API on all interfaces                | Partial (5/7)                            |
| HOST-01        | Med  | Password root SSH on some hosts           | Jarvis hardened; others via role (DR-12) |
| AI-01          | Med  | Unauthenticated LLM endpoint              | Mitigated (iptables lock)                |
| DETECT-01b/c/d | Info | Guest SIEM, adversary alerts, 2nd channel | Open                                     |

### Positive controls

No inbound exposure; L3 segmentation with default-deny and deny-logging; applied BMC egress clamp; authenticated cluster heartbeat; unprivileged containers; least-privilege API tokens for the dashboard and monitor; grade-A TLS; a working SIEM with dead-man alerts; an age-encrypted break-glass credential path independent of the cluster. For a single-operator home lab this is a strong, coherent posture. The material gaps are enforcement of the hardening role, rotation of the at-rest workstation credentials, and detection coverage of guests and the network edge.

## 13 Power and UPS Resilience

### 13.1 Live UPS state

Both UPS units were read live via NUT (upsc, E-07):

| UPS                                 | Status | Battery | Load | Runtime / feeds               |
|-------------------------------------|--------|---------|------|-------------------------------|
| Middle Atlantic UPS-OL2200R (UPS-B) | OL     | 100%    | 36%  | ~29 min; R730s, Randy, DS4246 |



Tripp Lite SMART1500VA OL 100% 59% ~9.5 min; EX3400, UniFi, EliteDesk nodes (UPS-A)

Both are online at full charge with clean input voltage (113-114 V). The concern is the imbalance: UPS-A (the network and core-services plane) sits near 60% load with under ten minutes of runtime, while UPS-B (the heavy servers) sits at 36% with nearly half an hour. This confirms the prior reliability finding (P-03 / R-18) and is logged as QL-R17.

## 13.2 Power budget

Estimated from live GPU draw (E-02/E-03) plus platform baselines. GPUs were near-idle at capture: QuarkyLab RTX 8000 at 3.7 W, Jarvis RTX 6000 pair at ~62 W and ~43 W.

| Load group                | Idle (est.) | Peak (est.)            | UPS |
|---------------------------|-------------|------------------------|-----|
| QuarkyLab R730 (idle GPU) | ~220 W      | ~650 W (GPU + CPU)     | B   |
| Jarvis R730 (2× RTX 6000) | ~300 W      | ~700 W (72B inference) | B   |
| Randy + DS4246 shelf      | ~300 W      | ~450 W                 | B   |
| 4× EliteDesk nodes        | ~120 W      | ~260 W                 | A   |
| EX3400 + UniFi + PDU      | ~120 W      | ~180 W                 | A   |
| <b>Estimated total</b>    | ~1060 W     | ~2240 W                | A+B |

These are engineering estimates, not metered values; only the GPU draw and UPS load percentages are measured. Adding the planned R730xd backup node would add roughly 150-350 W depending on transcode activity, and should land on UPS-B (or better, cross-fed) - but UPS-B has the headroom for it whereas UPS-A does not.

## 13.3 Scenarios

**Normal load:** both UPS comfortably within capacity. **Peak (GPU + inference):** UPS-B rises toward 55-60% and its runtime falls proportionally, still adequate for a graceful shutdown. **Extended outage:** UPS-A is the binding constraint - at ~9.5 min it must trigger a graceful shutdown of the network and core-services plane quickly, and DNS/routing will drop before the servers do. Rebalancing load onto UPS-B (QL-REC08) is the single highest- value power action.

## 13.4 NUT graceful shutdown and dual-PSU cross-feed

NUT monitors both UPS units (confirmed live via `upsc -l` on pve3). The open audit item R-07 remains: confirm `upsmon` runs on every node with a correct `SHUTDOWNCMD` and that staged `HOSTSYNC`/timers sequence a clean shutdown (storage last, edge first in the reverse of the recovery order). Equally important and not yet evidenced: the dual-PSU R730s should have one cord on each UPS. If both cords are on UPS-B, the redundant PSU buys nothing against a UPS-B fault. This should be physically verified.

## 13.5 Shutdown and startup sequencing

The safe shutdown order is the inverse of the recovery dependency graph (Section 17, diagram `recovery-dependencies.mmd`): shed non-critical guests, then app tier, then AI/research, then RKE2, then core services, then OPNsense, then storage (Randy last so backups and NFS stay available longest), then hypervisors. Startup reverses this: power and network, hypervisors, quorum, Randy storage, OPNsense, DNS, then up the stack. This sequencing should be captured as a one-page runbook; today it is implicit.

## 13.6 Efficiency

Two efficiency wins are already banked from the prior reliability work: CP-001 (GPU persistence / P0 idle fix on Jarvis, roughly -72 W) and CP-002 (CPU governors on pve3/4/5/Randy, roughly -90 W total). Remaining low-risk candidates: spinning down or powering off the DS4246 while bulk is near-empty (P-04, roughly -100 W - but weigh against the faulted-drive replacement work), and correcting the UPS-A/UPS-B imbalance to keep both units in their efficient load band. Consistent with the review's intent, none of these should be pursued at the expense of reliability; the drive replacement and load rebalance come first.



## 14 Documentation and Operational Readiness

The operative question for this phase is whether a competent administrator who has never seen NetFRAME could operate and recover it from the written record alone. The short answer is: for day-to-day operation and most single-service recovery, yes; for a full cold rebuild, mostly, provided they know that CLAUDE.md and the recent vault AARs supersede the front-facing documents.

### 14.1 Strengths

The documentation culture here is well above the home-lab norm and is a genuine portfolio asset.

- CLAUDE.md is the single most accurate document in the estate and is consistently ahead of every other source (E-12). It carries the authoritative node, IP, service, storage, and safety-note tables and reads as a living operations context file.
- A disciplined change record: roughly sixty dated runbooks and AARs under vault/Runbook/, ten Architecture Decision Records with context, decision, consequences, and alternatives, and formal post-incident RCAs (pve3 e1000e NIC hang, DHCP/Kea outage, backup-verify scheduler, Randy PCIe slot).
- Real disaster-recovery material: OPNsense serial-console DR with an age-encrypted config repository, a break-glass credential snapshot on both Ares and Randy that does not depend on the cluster, a node-unreachable triage guide, and a full laptop-recovery runbook (E-21).
- The Obsidian vault at Home-Lab/vault/ is declared the authoritative knowledge base and is structured into ADR, Compute, Infrastructure, Networking, and Runbook areas with a master map of content.
- Documentation-as-code with CI enforcement: a GitHub Actions job regenerates the topology Mermaid from topology/inventory.yml and fails the build if the drawing drifts from source.

### 14.2 Weaknesses

The dominant defect is **MEDIUM** one-directional drift (QL-R19): recent changes land in CLAUDE.md and the AARs but do not propagate outward to README.md, topology/, vault/00 - Homelab MOC.md, and the docs/ mirror.

- Stale clusters D-01 through D-10: Grafana still drawn on pve3 (moved to pve4), Headscale still on pve3 (moved to pve5), Randy CPU still v4/28c (corrected to v3/24c/48t), the DS4246 bulk pool still 58.2 T/2-vdev (expanded to 80 T/3-vdev), OPNsense still 25.7 in about twelve places (live is 25.1.12; 25.7 is the future CARP target), homepage still shown on pve1 CT104 (now pve3 CT106), and the EX3400 native-vlan-id note still wrong.
- Duplicate sources of truth: the same runbook lives in three homes (runbooks/, docs/, and vault/Runbook/) with real content drift in the Randy commissioning runbook and cosmetic drift elsewhere.
- Historical documents presented in the present tense: runbooks/EX3400-SSH-Auth-Failure-RCA.md (references the switch at .10.2, now the UniFi Dream Router) and homelab-setup.md need clear "historical/superseded" banners (D-10).

### 14.3 Documentation Gap Register

State key: **OK** well documented and current, **DRIFT** documented but stale in secondary sources, **THIN** exists but incomplete, **TRIBAL** lives mainly in the operator's head or chat history.

| Category                     | State | Recommended canonical source                                     |
|------------------------------|-------|------------------------------------------------------------------|
| Architecture / topology      | DRIFT | topology/ generated from inventory.yml; reconcile with CLAUDE.md |
| Physical inventory / asset   | THIN  | Asset register (this review, Appendix B) + CLAUDE.md tables      |
| Rack layout                  | THIN  | vault/Rack Layout.md (add rack SVG, open loop)                   |
| IP address management (IPAM) | OK    | CLAUDE.md network tables; consider a dedicated IPAM note         |
| VLAN / trunk design          | DRIFT | vault/Networking/ + EX3400 config once captured (QL-REC09)       |
| Firewall policy              | OK    | vault/Runbook/Security-VLAN-Segmentation-*                       |
| DNS / DHCP                   | OK    | CLAUDE.md safety notes + DNS-HA runbook                          |
| Proxmox / virtualization     | OK    | CLAUDE.md + Cluster-Update-Procedure                             |
| Storage                      | DRIFT | CLAUDE.md Storage section (docs mirror lags)                     |
| Backup                       | OK    | backup-verify role + this review Appendix E                      |
| Kubernetes / RKE2            | OK    | RKE2 phase runbooks                                              |
| AI services / Jarvis         | OK    | netframe-monitor/constitution/ + jarvis-ai-ops .tex              |
| Monitoring / alerting        | OK    | netframe-monitoring-stack repo (config-as-code)                  |
| UPS / power                  | THIN  | vault/Power Distribution.md + this review Section 13             |
| Startup / shutdown sequence  | THIN  | Formalize from recovery-dependencies diagram + NUT plan          |
| Incident response            | OK    | AAR corpus                                                       |
| Disaster recovery            | OK    | OPNsense DR, break-glass, laptop-recovery                        |
| User access                  | OK    | QuarkyLab account onboarding runbook                             |

| Category                         | State  | Recommended canonical source                              |
|----------------------------------|--------|-----------------------------------------------------------|
| Student onboarding / offboarding | THIN   | add-cluster-key.sh; offboarding automation planned        |
| Scientific workload deploy       | THIN   | Researcher guide (.tex) + SLURM phase docs                |
| Change management                | OK     | Manual-deploy policy + CI/CD runbook                      |
| Secret rotation                  | TRIBAL | Vaultwarden is the store; rotation procedure undocumented |
| Certificate renewal              | THIN   | step-ca; only registry cert monitored (QL-REC07)          |
| Firmware / lifecycle             | THIN   | Per-node BIOS/iDRAC notes scattered; consolidate          |
| Hardware replacement             | OK     | Randy PCIe slot + storage expansion runbooks              |

## 14.4 Recommendation

Adopt a single-source generation model for the docs/ tree so that the vault (or CLAUDE.md) is the sole authority and docs/ and runbooks/ are generated or reduced to pointers (QL-REC11). The immediate action is to land the D-01 through D-10 documentation reconciliation, which is already staged on the audit/2026-07-22-reconciliation branch, flowing corrections from CLAUDE.md outward and keeping the 25.7 reference only in the CARP HA plan. Secret-rotation and certificate-renewal procedures are the two categories most reliant on tribal knowledge and should be written down next.

## 15 Scientific Research and Student Support

QuarkyLab is a genuinely multi-tenant platform: it carries a production neutrino-physics ML workload, a shared teaching environment for computer-science students, and supporting AI tooling, all on one RTX 8000. This section separates what works as a home-lab convenience from what would be required for a formally governed student and research service.

### 15.1 Research workloads

The physicist fernanda runs a production DUNE reconstruction ML workload on the RTX 8000 (live-confirmed present and idle at review time, E-02). A DUNE Agent RAG pipeline over the dunereco codebase assists onboarding, and a second researcher builds a separate RAG pipeline (ARID) under a scoped exemption. Research data is protected from student starvation by a refreservation=2T floor on the researcher dataset.

### 15.2 Multi-tenant GPU sharing (live-validated 2026-07-02)

One physical Turing card (no MIG) is shared safely between research and roughly 15 to 20 students per semester:

- **Scheduling.** SLURM gres/shard with 8 shards allows up to 8 concurrent GPU jobs while leaving the card in Default compute mode so the researcher runs native.
- **Hard VRAM cap.** A per-job NVIDIA MPS instance inside each student container enforces a per-job VRAM limit (about 6 GB per shard), chosen over SLURM gres/mps to avoid forcing exclusive-process mode on the researcher.
- **Isolation.** job\_submit.lua wraps every student job in aptainer exec --nv --network=none with a private home, read-only shared data, and a RAM bound; cgroup.conf ConstrainDevices=yes denies /dev/nvidia\* to any job without a GPU GRES.
- **Priority.** The research partition (PriorityTier=100) preempts student jobs (PreemptMode=REQUEUE, not suspend, so freed VRAM is actually released), and a queued job auto-restarts.
- **Fairness.** Multifactor fairshare with a 3-day decay favors students who have used the GPU least; studentqos caps one running job per student; students are batch-only.

The Phase 04 validation table shows every case passing: no-GRES jobs denied the device, a student request for the whole card rejected, VRAM out-of-memory at the expected cap, 3 then 8 concurrent students, the researcher native at 47.8 GB, and preemption reclaiming the card.

### 15.3 Reproducibility and access lifecycle

Reproducibility rests on a curated Aptainer image (base.sif, pinned torch and TensorFlow) run from a local copy and refreshed daily so a storage hiccup cannot block a launch. Access is key-only SSH over a Cloudflare Tunnel with no inbound ports; onboarding is a scripted add-cluster-key.sh that validates the key and creates a locked-down account, and a group sshd policy denies password auth, all forwarding, and sudo. Offboarding removes the tailnet node and the user account.

## 15.4 Readiness assessment

| Strength (production-grade)                              | Gap for a governed service                          |
|----------------------------------------------------------|-----------------------------------------------------|
| Real device isolation via cgroups + app-tainer + MPS cap | No per-student network isolation within the node    |
| Research preemption and fairshare                        | Research and student data both single-copy (QL-R02) |
| Per-user ZFS quotas and read-only shared data            | No formal data-retention or privacy policy          |
| Reproducible pinned container images                     | Offboarding not yet automated (planned)             |
| Batch-only, no interactive un-capped access              | Student containers not covered by Wazuh (QL-R22)    |

### Verdict

For a home lab this is exceptional: the GPU multi-tenancy is designed and validated to a standard most small HPC groups would accept, and the research floor plus preemption model is a real answer to the hardest problem in shared GPU teaching. The gaps are governance rather than mechanism: a second copy of research data (QL-R02, QL-REC03), a written retention and privacy posture, automated account lifecycle, and extending SIEM coverage to student workloads would move this from a well-run lab to a defensible institutional service.

## 16 Enterprise and Fortune 100 Alignment

This comparison is deliberately about capability and operational discipline, not budget. The lab is not penalized for lacking commercial products where an open-source practice delivers the same control. Each dimension is classified as **Comparable** to mature enterprise practice, **Partially aligned**, **Developing**, **Material gap**, or **N/A at this scale**.

| Dimension                | Classification    | Evidence and realistic improvement                                                                                                       |
|--------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Asset management         | Partially aligned | CLAUDE.md tables + this review's register; no live CMDB. Improve: keep the asset register generated.                                     |
| Configuration management | Partially aligned | Ansible hardening + backup_verify roles, lint-clean; but hardening is --check only, not enforced (DR-12). Improve: apply and gate drift. |
| Network segmentation     | Comparable        | Seven VLANs, default-deny inter-VLAN, applied BMC egress clamp, VLAN30 to VLAN1 clamp, deny-logging. A genuine strength.                 |
| Identity management      | Developing        | Shared root SSH, no MFA, single operator. Improve: per-admin accounts, MFA on Vaultwarden/Grafana.                                       |
| Secrets management       | Partially aligned | Vaultwarden + offline break-glass age snapshot. Gap: plaintext creds at rest on Ares (QL-R07), no rotation procedure.                    |
| Monitoring               | Partially aligned | Prometheus/Grafana/Loki, 15/15 targets, dead-man heartbeats. Gaps: single-node stack (QL-R10), missing PVE/cert rules (QL-REC06/07).     |
| Logging                  | Partially aligned | Loki + OPNsense/EX3400 syslog + Wazuh on hosts. Gaps: Loki unauthenticated, guests not in SIEM (QL-R22).                                 |
| Incident management      | Comparable        | Formal RCAs and AARs with rollbacks; better than many small enterprises.                                                                 |
| Change management        | Partially aligned | Manual-deploy-only policy, CI lint/branch-protection, pre-push hooks. Gap: no formal change calendar/approvals beyond self.              |
| Backup                   | Partially aligned | PBS nightly, verified, fresh (E-08). Gap: 3-2-1 not met (QL-R03), pve1 and bulk uncovered (QL-R06/R02).                                  |
| Disaster recovery        | Partially aligned | DR runbooks, break-glass, restic restore drill done. Gap: no full cluster-loss rehearsal; single site.                                   |
| Capacity planning        | Partially aligned | Deterministic forecast in netframe_predict.py; GPU/quota dashboards. Gap: no formal capacity model.                                      |
| Security review          | Comparable        | Multiple prior assessments (enterprise, reliability, security, red-team) with tracked remediation. Unusually mature.                     |
| Vulnerability management | Developing        | Patch lag 63-142/node (QL-R13); no formal CVE tracking cadence. Improve: staged patch windows (QL-REC19).                                |
| Documentation            | Partially aligned | Rich but drifting (Section 14).                                                                                                          |
| Automation               | Partially aligned | Ansible, systemd timers, CI, AI-ops. Gap: infra deploy stays manual by policy (defensible).                                              |
| Testing                  | Partially aligned | netlab containerlab CI boots real routers and runs reachability/failover tests. Rare and impressive.                                     |
| Release management       | Developing        | Config-as-code repos, but no versioned release process for infra.                                                                        |
| Infrastructure as code   | Partially aligned | Ansible + config-as-code monitoring; not fully declarative (no Terraform for PVE).                                                       |
| Service ownership        | Developing        | Single operator owns everything; clear but bus-factor 1.                                                                                 |

| Dimension                   | Classification | Evidence and realistic improvement                                                                   |
|-----------------------------|----------------|------------------------------------------------------------------------------------------------------|
| SLIs / SLOs / error budgets | Material gap   | Not defined. Improve: define a few SLOs (DNS availability, backup freshness) - high portfolio value. |
| Dependency mapping          | Comparable     | This review plus the dependency and recovery diagrams; netframe_monitor computes blast radius.       |
| Business continuity         | Developing     | Break-glass + DR docs; single-site, single-operator constrain true BC.                               |
| Vendor lifecycle            | Developing     | EOL R730/EX3400/Mac-Mini tracked informally; no formal lifecycle register.                           |
| Auditability                | Comparable     | Hash-chained tamper-evident AI ledger, full git history, this audit trail.                           |
| Risk acceptance             | Comparable     | Risks explicitly accepted and documented (pinned kernels, raidz1, single-node monitoring).           |

## 16.1 What Makes This Unusually Strong for a Home Lab

NetFRAME runs three real, non-trivial production workloads (DUNE neutrino-physics research, a multi-tenant student GPU teaching cluster, and a self-hosted LLM inference platform) on properly segmented infrastructure. Defense-in-depth is not theoretical here: BMC isolation with an applied egress clamp, default-deny inter-VLAN policy, and least-privilege monitoring credentials are all live. The AI-operations system is safety-first by design (Section 10), with a deterministic evidence engine and human-gated remediation that would pass a serious enterprise review. Above all, the lab audits itself honestly (this review sits alongside four prior assessments and an in-flight reconciliation branch), which is the single strongest signal of operational maturity.

## 16.2 What Would Concern an Enterprise Hiring Manager

Bus factor of one: every design, credential, and recovery path depends on a single operator. There are no defined SLOs or error budgets, so reliability is asserted rather than measured. The environment is single-site with backups still short of 3-2-1 (QL-R03), and a meaningful set of security remediations remain open or blocked (registry auth, NetworkPolicies, guest SIEM coverage). Documentation drift means the public-facing docs would mislead a new hire until the reconciliation lands.

## 16.3 Portfolio Framing

For a candidate transitioning into network and infrastructure engineering, this environment demonstrates enterprise judgment far beyond the typical home lab: real segmentation, real DR rehearsal, real capacity and security review, and honest risk acceptance. The highest-leverage additions for recruiter and hiring-manager impact are defining two or three SLOs, closing the 3-2-1 gap, and landing the documentation reconciliation so the public repo reads as cleanly as the operator knows the estate to be.

# 17 Failure-Mode and Dependency Analysis

This section walks the credible failure scenarios for the estate. It is analytical only: **no failure injection was performed**, consistent with the read-only mandate. The dependency structure it draws on is in `diagrams/service-dependencies.mmd` and the recovery order in `diagrams/recovery-dependencies.mmd`.

The single most important structural observation is a small set of keystone dependencies: OPNsense (routing/DHCP/DNS egress), Randy (all backups, NFS, and k8s storage), the single Unbound resolver, and the co-located observability stack. Most high-impact scenarios below trace back to one of these four.

| Scenario            | Detection                  | User impact                                  | Response / mitigation gap                                                       | Safe test?            |
|---------------------|----------------------------|----------------------------------------------|---------------------------------------------------------------------------------|-----------------------|
| Loss of 1 PVE node  | InstanceDown, quorum votes | guests on that node down until manual action | Quorum holds (4/7+); no HA groups so restart guests elsewhere manually (QL-R18) | Yes (drain in window) |
| Loss of 2 PVE nodes | quorum + InstanceDown      | possible quorum loss depending on nodes      | 5/7 still quorate; below 4 votes cluster read-only; recover a node              | No                    |
| Loss of quorum      | ClusterNotQuorate          | config read-only, no guest changes           | Restore a node or set expected votes; documented                                | No                    |

| Scenario                         | Detection                              | User impact                                           | Response / mitigation gap                                                     | Safe test?             |
|----------------------------------|----------------------------------------|-------------------------------------------------------|-------------------------------------------------------------------------------|------------------------|
| Loss of OPNsense (QL-R04)        | InstanceDown, WAN alerts, user reports | whole LAN: no routing/DHCP/egress                     | Serial console DR + age config restore; CARP HA planned (QL-REC10)            | No                     |
| Loss of DNS                      | Pihole probes                          | name resolution fails for single-resolver clients     | Dual Pi-hole HA + both handed out; single Unbound is the residual risk (R-03) | Yes (fail one Pi-hole) |
| Loss of DHCP                     | user reports, Kea health               | new leases fail; static-in-container svcs unaffected  | Kea on OPNsense; critical hosts static; restore OPNsense                      | Partial                |
| Loss of EX3400                   | everything dark, no alert egress       | full L2 outage                                        | No switch redundancy; config not captured (QL-R16); manual rebuild            | No                     |
| Loss of a trunk                  | link/port down (if monitored)          | VLAN reachability degraded                            | Redundant UniFi path intact (xe-0/2/3 confirmed up 10G, DR-14 resolved)       | Yes (in window)        |
| Loss of a corosync link          | corosync ring status                   | ring degraded, quorum may wobble                      | knet single-link per node; ring on stable VLAN 1                              | No                     |
| Loss of Randy / storage (QL-R05) | InstanceDown, NFS/PBS errors           | backups, NFS /data, k8s PVs all gone at once          | Second target planned (R730xd); no live failover today                        | No                     |
| Loss of DS4246                   | zpool import fail, bulk offline        | bulk data (research+media) unavailable                | Single enclosure SPOF; bulk has no second copy (QL-R02)                       | No                     |
| Loss of Jarvis                   | NetframeMonitorStale, llm down         | LLM inference + AI-ops reporting down                 | Deterministic alerter still runs; non-critical to core infra                  | Yes (stop svc)         |
| Loss of Quarky-Lab               | InstanceDown, SLURM down               | research + student GPU + Wazuh down                   | Wazuh migratable; workspace pool local; students queue                        | No                     |
| Loss of Grafana (QL-R10)         | dead-man heartbeat                     | no metrics/logs/alert evaluation                      | Stack co-located on CT103; heartbeat can catch total loss                     | Yes (in window)        |
| Loss of Headscale                | tailnet coordination fails             | admin mesh only; no user impact                       | Existing nodes keep working; rebuild from backup                              | Yes                    |
| Loss of NPM                      | reverse-proxy 502s                     | public hostnames down; internal IPs still work        | Restore CT101; static backends reachable directly                             | Yes                    |
| Loss of internet                 | WAN alerts                             | no egress; alerting via Discord also blinded (QL-R09) | WAN2 5G failover; add 2nd OOB alert channel (QL-REC14)                        | Yes (unplug WAN1)      |
| Loss of 1 UPS (QL-R17)           | UPS status via NUT                     | that plane loses ride-through                         | UPS-A at 9.5 min is the tighter margin; rebalance (QL-REC08)                  | Yes (on battery test)  |
| Extended power outage            | UPS runtime alerts                     | staged shutdown then full stop                        | NUT graceful shutdown to audit (R-07); startup order documented               | No (real outage)       |
| Battery failure                  | UPS self-test/charge alerts            | no ride-through on that UPS                           | Both 100% now; batteries age unknown; schedule test                           | No (needs approval)    |
| Certificate expiry               | RegistryCertExpiring (registry only)   | TLS breaks for unmonitored certs                      | Broaden cert monitoring (QL-REC07); step-ca renewal                           | Yes                    |
| GitHub unavailable               | push/pull fails                        | no config-as-code sync; infra unaffected              | Local clones remain; manual deploy already the norm                           | Yes                    |

| Scenario                      | Detection                   | User impact                                 | Response / mitigation gap                                               | Safe test?         |
|-------------------------------|-----------------------------|---------------------------------------------|-------------------------------------------------------------------------|--------------------|
| Credential loss (QL-R08)      | operator-detected           | cannot unlock services                      | Break-glass age snapshot on Ares+Randy (offline)                        | Yes (test decrypt) |
| Accidental config deletion    | drift/audit, user reports   | service or rule removed                     | Git history + PBS + OPNsense config backup; restore                     | Yes (in test)      |
| Failed k8s/PVE upgrade        | node/pod failures           | control plane or node degraded              | Rolling upgrades keep quorum; kernel holds protect GPU nodes (QL-REC19) | Partial            |
| Compromised student account   | Wazuh/SLURM anomalies       | contained by app-tainer + cgroups + no sudo | Batch-only, network=none, VRAM caps; guest SIEM blind spot (QL-R22)     | Yes (test acct)    |
| AI hallucination / unsafe rec | policy screen, audit ledger | none: no autonomous action possible         | Deterministic evidence engine + human-gated allowlist remediation       | Yes (eval harness) |
| Monitoring failure (QL-R11)   | dead-man heartbeat          | silent blind spot; a real risk here         | ZfsPoolDegraded delivery unverified; verify path (QL-REC02)             | Yes (synthetic)    |
| Backup corruption             | PBS verify failures         | restore may fail                            | Weekly verify + GC; add stale/verify alerts (QL-REC07)                  | Yes (verify job)   |

Several of these are safely testable in production with a maintenance window (fail one Pi-hole, unplug WAN1 to exercise 5G failover, drain a node, restore a guest to a scratch VMID, test-decrypt break-glass). The genuinely unsafe tests - dual-node loss, storage loss, real power outage, battery failure - should only be modeled, never injected.

## 18 Drift and Contradiction Register

This phase compares five views of the estate: the state *declared* by the review specification, the state *configured* on devices, the state *observed* live on 2026-07-23, the state *documented* in the repository, and the *most likely correct* value adjudicated from the strongest evidence. No drift was resolved during this read-only review; each row records the recommended resolution and whether operator approval is required. The full field set is in `data/drift-register.csv`.

Two of the review specification's own "known corrections" are inverted versus live reality and must **not** be applied: DR-01 (GPU assignment) and DR-02 (OPNsense version direction). Live evidence and the repository agree against the specification on both.

| ID    | Item                  | Observed / configured                              | Documented                           | Correct                | Appr.   |
|-------|-----------------------|----------------------------------------------------|--------------------------------------|------------------------|---------|
| DR-01 | GPU assignment        | QuarkyLab RTX 8000; Jarvis 2x RTX 6000 (E-02/E-03) | CLAUDE.md correct; 1 stale vault log | Live/repo              | No      |
| DR-02 | OPNsense version      | Not re-verified live (E-23)                        | 25.1.12 vs 12 refs at 25.7           | 25.1.12                | No      |
| DR-03 | Grafana node          | CT103 on pve4 (E-06)                               | README/topology say pve3             | pve4                   | No      |
| DR-04 | Headscale node        | CT105 on pve5 (E-06)                               | README/ADR say pve3                  | pve5                   | No      |
| DR-05 | Randy CPU             | E5-2690 v3 24c/48t                                 | docs say v4 28c/56t                  | v3                     | No      |
| DR-06 | Randy Seagate drives  | 4 in raidz2-3 (E-06)                               | docs say 2 spare                     | 4 in-pool              | No      |
| DR-07 | bulk geometry         | 3x RAIDZ2 80 T (E-04)                              | 2x8-wide 58.2 T                      | 3-vdev 80 T            | No      |
| DR-08 | EX3400 native-vlan-id | <b>Verified live</b> in config (E-25)              | docs say unsupported                 | Supported (ELS)        | No      |
| DR-09 | Homepage location     | CT106 on pve3 (E-06)                               | pve1 CT104                           | pve3 CT106             | No      |
| DR-10 | Randy PVE version     | <b>9.2.3</b> live (E-04)                           | docs say 9.1.1                       | 9.2.3                  | No      |
| DR-11 | pve1 CT104 homepage   | <b>running</b> (E-05)                              | docs say retired 06-24               | stop/disable           | Yes     |
| DR-12 | Ansible hardening     | -- check only (E-24)                               | role exists                          | not enforced           | Yes     |
| DR-13 | DS4246 cabling        | Not live-inspected                                 | repo vs spec differ                  | verify at HW           | No      |
| DR-14 | xe-0/2/3 SFP link     | <b>UP at 10G</b> (E-22)                            | recorded DOWN (EEPROM)               | recorded fault         | No      |
| DR-15 | EX2300 lab switch     | <b>Does not exist</b> (E-22)                       | listed in docs/inventory             | is stale phantom asset | No      |
| DR-16 | EX3400 STP root       | UniFi was root; now EX3400 (E-28)                  | docs claimed 4096 root               | remediated             | Applied |



| ID    | Item                  | Observed / configured    | Documented     | Correct              | Appr.    |
|-------|-----------------------|--------------------------|----------------|----------------------|----------|
| DR-17 | ZFS collector version | live had capacity metric | repo lacked it | reconciled (d2858df) | Resolved |

## 18.1 Interpretation

DR-03 through DR-09 are the classic one-directional documentation drift: an unambiguous, doc-recorded canonical value exists and the fix is doc-only, already staged on the reconciliation branch. DR-01 and DR-02 are the reverse trap, where the incoming specification is stale and applying its "correction" would introduce error; live `nvidia-smi` (E-02/E-03) and the operator's own AAR settle both.

Four items are newly surfaced by this review's live pass rather than the prior repo audit. DR-10 (Randy running PVE 9.2.3 while every document still says 9.1.1) is a benign but real configuration-versus-documentation gap and should be reconciled after confirming the upgrade was intended. DR-11 (the retired homepage container CT104 still running on pve1) and DR-12 (the hardening role never actually applied to nodes, only - - check verified) are drift between assumed and actual runtime state and both require approval to resolve because they are service- or configuration-affecting. DR-14 (the UniFi SFP uplink down on an EEPROM speed mismatch) is a lost redundant path needing a hardware change.

## 19 Risk Register

Severity follows a fixed model and is deliberately not inflated. **CRITICAL** an immediate, credible risk of catastrophic outage, unrecoverable loss, or serious compromise. **HIGH** a significant risk requiring prompt corrective planning. **MEDIUM** a material weakness to schedule. **LOW** an improvement opportunity or limited risk. **INFO** a documentation or visibility observation. Every risk is tied to environment evidence; the full field set (description, existing controls, residual risk, validation method, confidence) is in `data/risk-register.csv`.

### 19.1 Summary

**0 Critical, 7 High, 12 Medium, 3 Low.** No finding meets the Critical bar today: the most dangerous condition (a faulted drive in the bulk pool) is still tolerated by RAIDZ2 with no data errors and a replacement in progress. It is rated High rather than Critical, but it is the top operational priority because QL-R01 pairs with QL-R02 (the bulk pool has no second copy) to form the single most serious combined exposure in the estate: one more drive failure in the same vdev before replacement would mean permanent loss of data that exists nowhere else.

### 19.2 High

| ID     | Title                                        | Category   | Treatment                                 |
|--------|----------------------------------------------|------------|-------------------------------------------|
| QL-R01 | Faulted bulk drive in a degraded RAIDZ2 vdev | Storage    | Replace mpathv + scrub (QL-REC01)         |
| QL-R02 | bulk pool has no second copy                 | Backup     | Add bulk/fernanda to PBS or B2 (QL-REC03) |
| QL-R03 | All backups on one node/rack/UPS (3-2-1 gap) | Backup/DR  | Second target R730xd or B2 (QL-REC03)     |
| QL-R04 | OPNsense is a single point of failure        | Network    | CARP HA pair (QL-REC10)                   |
| QL-R05 | Randy is a single storage node               | Storage    | Second storage/backup target              |
| QL-R11 | ZfsPoolDegraded alert delivery unverified    | Monitoring | Verify Discord delivery (QL-REC02)        |

QL-R01 through QL-R05 and QL-R11 are the High tier. QL-R11 is rated High because a silent critical alert is more dangerous than no alert: the degraded pool was in fact surfaced by a staleness alert, not by the pool-health rule, so the delivery path is unproven.

### 19.3 Medium

| ID     | Title                                | Category    | Treatment                       |
|--------|--------------------------------------|-------------|---------------------------------|
| QL-R06 | pve1 guests unprotected (no backup)  | Backup      | vzdump pve1 to Randy (QL-REC04) |
| QL-R07 | GitHub PAT plaintext at rest on Ares | Security    | Rotate + move to env (QL-REC05) |
| QL-R08 | Secrets-recovery circular dependency | Security/DR | Keep break-glass current + test |

| ID     | Title                                    | Category    | Treatment                            |
|--------|------------------------------------------|-------------|--------------------------------------|
| QL-R09 | Single WAN/Discord alert path            | Monitoring  | Add out-of-band channel (QL-REC14)   |
| QL-R10 | Observability stack co-located on one CT | Monitoring  | External alert eval / heartbeat      |
| QL-R12 | Kernel/GPU hold asymmetry on Jarvis      | Reliability | Mirror QuarkyLab holds (QL-REC12)    |
| QL-R13 | Fleet patch lag (63-142/node)            | Security    | Staged patch + reboots (QL-REC19)    |
| QL-R14 | Unauthenticated LAN AI/service endpoints | Security    | Registry auth; keep locks (QL-REC15) |
| QL-R15 | No NetworkPolicies / PSA in RKE2         | Security    | Default-deny + PSA (QL-REC15)        |
| QL-R16 | EX3400 config not captured; SSH broken   | Network/DR  | Reconcile SSH + Oxidized (QL-REC09)  |
| QL-R17 | UPS-A overloaded / short runtime         | Power       | Rebalance loads (QL-REC08)           |
| QL-R22 | Wazuh coverage limited to hypervisors    | Security    | Guest agents + detections            |

## 19.4 Low and Accepted

| ID     | Title                                    | Category      | Disposition                       |
|--------|------------------------------------------|---------------|-----------------------------------|
| QL-R18 | HA fencing in standby (no auto recovery) | Availability  | Decide per guest; may be accepted |
| QL-R20 | Randy PVE version drift (9.2.3 vs docs)  | Configuration | Doc reconcile (Low)               |
| QL-R21 | raidz1 on aged disks                     | Storage       | Accepted; monitor SMART, spares   |

Every entry above traces to a cited evidence item and, where verifiable, was confirmed by at least two independent read-only methods (for example QL-R01 by live `zpool status`, the prior audit capture, and the DS4246 slot-15 incident record). Confidence per risk is recorded in the CSV; the two items carrying Medium confidence are QL-R05 and QL-R11, both constrained by what a read-only pass can prove.

## 20 Recommendations

Every recommendation below is tied to specific evidence from this environment. A finding is not automatically a recommendation, and stable systems are not changed without a compelling reason. This review executed **none** of these actions; all await explicit operator approval. The full machine-readable set with rollback, validation, effort, and impact fields is in `data/recommendation-register.csv`.

### 20.1 By Class

| ID       | Class               | Action                                    | Approval  |
|----------|---------------------|-------------------------------------------|-----------|
| QL-REC02 | Immediate follow-up | Verify ZfsPoolDegraded reached Discord    | No (read) |
| QL-REC11 | Immediate follow-up | Land the D-01..D-10 documentation PR      | No (docs) |
| QL-REC01 | Urgent corrective   | Replace faulted bulk drive + scrub        | Yes       |
| QL-REC05 | Urgent corrective   | Rotate at-rest creds on Ares              | Yes       |
| QL-REC04 | Next window         | Back up pve1 guests to Randy              | Yes       |
| QL-REC08 | Next window         | Rebalance UPS loads + verify NUT shutdown | Yes       |
| QL-REC12 | Next window         | Mirror Jarvis GPU package holds           | Yes       |
| QL-REC19 | Next window         | Staged fleet patch + reboots              | Yes       |
| QL-REC03 | Near-term           | Second backup copy (R730xd or B2)         | Yes       |
| QL-REC06 | Near-term           | Deploy prometheus-pve-exporter            | Yes       |
| QL-REC07 | Near-term           | Add quorum/backup-stale/verify/cert rules | Yes       |
| QL-REC09 | Near-term           | Reconcile EX3400 SSH + deploy Oxidized    | Yes       |
| QL-REC14 | Near-term           | Second out-of-band alert channel          | No        |
| QL-REC18 | Near-term           | .gitignore in 6 repos + OPSEC review      | No        |
| QL-REC10 | Medium-term         | Build OPNsense CARP HA pair               | Yes       |
| QL-REC13 | Medium-term         | Wire remediation gate to evidence engine  | No        |
| QL-REC15 | Medium-term         | Registry auth + RKE2 NetworkPolicies      | No        |
| QL-REC16 | Optional            | Mask failed openipmi.service on pve1      | Yes       |
| QL-REC17 | Optional            | Confirm + retire duplicate homepage CT104 | Yes       |
| QL-REC20 | No action           | Keep pinned GPU kernels (do not upgrade)  | No        |
| -        | No action           | Do not enforce VLAN-1 egress lockdown yet | No        |

## 20.2 Detailed Treatment of the Top Items

### QL-REC01 - Replace the faulted bulk drive (Urgent)

**Problem:** mpathv is FAULTED in raidz2-2; that vdev now has one of two parity drives (QL-R01, E-04). **Action:** insert a  $\geq 4$  TB SAS drive into one of the two free DS4246 bays, then `zpool replace bulk mpathv <new-by-id>`, watch the resilver, and scrub after. **Prerequisite:** a spare drive; identify the bay for WWN 5000c500631a54fb. **Validation:** `zpool status` returns ONLINE and a clean scrub. **Rollback:** pull the new drive; the pool returns to its current degraded state, not worse. **Window:** none needed. **Confidence:** High.

### QL-REC03 - Add a second backup copy (Near-term)

**Problem:** every backup lives only on Randy; bulk research data has no copy at all (QL-R02/R03). **Action:** PBS remote-sync to the planned R730xd second PBS, or `rclone/PBS` to Backblaze B2 for datastore plus bulk/fernanda. **Prerequisite:** the R730xd build or a B2 account. **Validation:** a test restore from the second copy. **Rollback:** remove the sync job. **Approval:** required. **Confidence:** High.

### QL-REC05 - Rotate at-rest credentials on Ares (Urgent)

**Problem:** a live GitHub PAT sits in plaintext in `/github_helper.py`, plus `/apikey.txt` and `/.wazuh-api-newpw.txt` (QL-R07, E-17). None are committed, but all are usable if the workstation is compromised. **Action:** rotate the PAT, move it and the other secrets to an environment file or credential store. **Validation:** the old token is revoked and scripts read from the environment. **Rollback:** retain the old file until the new path is confirmed. **Approval:** required (touches GitHub/Wazuh). **Confidence:** High.

### QL-REC10 - OPNsense CARP HA pair (Medium-term)

**Problem:** a single OPNsense VM on pve2 is a whole-network single point of failure (QL-R04). **Action:** follow the existing 2026-07-22 CARP plan: a second node, both on 25.7, a dedicated pfsync link, Kea DHCP HA, and a standalone backup host outside the cluster; the WAN-side CARP behavior is the hard open item to solve first. **Validation:** a failover test in a maintenance window. **Rollback:** documented in the plan. **Window:** required. **Confidence:** Medium.

## 20.3 Explicitly No Action

Keep the GPU-node kernels pinned at 6.14.11-9-pve; upgrading breaks the NVIDIA 550 stack (QL-REC20). Do not enforce the VLAN-1 egress lockdown yet: its prerequisites (UniFi discovery and multicast, RKE2 and container-registry outbound, CDN dependencies, administrative access paths, DHCP reservations, emergency access, current firewall API permissions, and a tested rollback) are not all satisfied, and this is a standing operator decision that this review respects.

## 20.4 Approval-Driven Next Steps

The recommended sequence, each pending operator sign-off, is: (1) the two immediate read-only follow-ups (QL-REC02 verify alert delivery, QL-REC11 land the docs PR); (2) the urgent corrective pair (QL-REC01 drive replacement, QL-REC05 credential rotation); (3) the next-window batch (QL-REC04 pve1 backup, QL-REC08 UPS rebalance, QL-REC12 Jarvis holds, QL-REC19 staged patching); then (4) the near- and medium-term engineering items led by QL-REC03 (second backup copy) and QL-REC10 (OPNsense HA). Approval for any one action does not authorize the others.

## 21 Limitations and Confidence

This was a read-only review conducted in a single working session. Its conclusions are only as strong as the read-only evidence that supports them, and several areas could not be examined to the depth a change-authorized or longer engagement would allow. The limitations below are stated plainly so that the reader can calibrate each conclusion.

## 21.1 Access and method limitations

- **Read-only ceiling.** No configuration-changing or service-affecting command was run, so any finding that would require mutating state to confirm (for example, an actual restore test, a failover, or a UPS self-test) is assessed from configuration and documentation rather than demonstrated behavior.
- **OPNsense not live-inspected (E-23).** By design, no OPNsense API credential was used and VM 100 was not touched. Firewall rules, the running version, DHCP reservations, and gateway status therefore rest on the repository and the operator's AARs. The version 25.1.12 is documented but not re-verified live; confidence in firewall specifics is **medium**.
- **EX3400 now live-inspected (E-22 / E-25) - limitation lifted.** SSH key authentication to the switch is still not configured, but the operator supplied password credentials during the review, so config, VLAN membership, spanning-tree state, optics, and neighbors were all pulled read-only. Switch facts are now **high** confidence. The residual gap is automation, not visibility: there is still no nightly config capture (QL-R16, QL-REC09).
- **Grafana rule load not independently confirmed.** The device-level ZfsDeviceFaulted rule was authored and the new metric was verified live in Prometheus, but the reviewer could not independently confirm that Grafana loaded the rule: the CT103 exec path repeatedly timed out and the alert API requires authentication. The end-to-end proof is operator confirmation that the alert reached Discord. The underlying metric is verified; only the rule-load step rests on the operator's report.
- **Faulted-drive identity.** Live multipath -ll mpathv returned empty during this session (E-04); the drive's WWN (5000c500631a54fb) and model (Seagate ST4000NM0023) are taken from the prior audit, which is corroborated by the live FAULTED state and the NF-INC-2026-07-22 record. Confidence in the fault is **high**; in the exact device identity, **medium-high**.
- **Binary artifacts not text-scanned.** Roughly 50 committed PDFs (concentrated in the assessment repositories) were not decoded for embedded secrets. The identity/secrets and IPAM PDFs in the enterprise-assessment repo warrant a human eyes-on for sensitivity (E-17).
- **Point-in-time metrics.** Live CPU, memory, UPS load, and job-queue readings are instantaneous. UPS runtime in particular is a load-dependent estimate and will vary; the imbalance conclusion (QL-R17) is robust, but the exact minutes are a snapshot.
- **Prior-assessment age.** The enterprise (E-18) and reliability (E-20) assessments are roughly 9 to 10 days old; some of their items have since been remediated (for example the GPU-persistence power fix). Their findings are used as corroboration and history, reconciled against current state where possible.

## 21.2 Confidence by major conclusion

- **High:** cluster quorum and membership; GPU assignment; ZFS pool health including the bulk fault; backup freshness; UPS online state and the load imbalance; repository secrets-cleanliness of tracked content; the at-rest workstation credential findings.
- **Medium-high:** the faulted device's precise identity; storage geometry details on the DS4246; the AI-ops safety controls (read from code, not runtime-traced).
- **Medium:** network control-plane specifics (OPNsense and EX3400); alert-rule delivery; the true applied state of the Ansible hardening role, which is --check-only and not enforced on nodes (E-24 / DR-12).

No unauthorized change was made at any point during this review, and no discovered credential was used or reproduced in full.

## A Command and Activity Log

The discovery and verification phases of this review were entirely read-only and low-risk. Later in the engagement the operator explicitly approved two corrective changes; those are the only entries below marked **Modified: Yes**, and each was verified afterwards by the reviewer. No destructive or external-target command was executed at any point. Timestamps are 2026-07-23 (EDT/UTC mixed as captured). The full raw output is retained in the reviewer session, not in this repository, to avoid committing transient host detail. The captured switch configuration is deliberately *not* committed because it contains the SNMP community string.

| Time  | System       | Command (summarized)                                                  | Purpose                                              | Result                                                              |
|-------|--------------|-----------------------------------------------------------------------|------------------------------------------------------|---------------------------------------------------------------------|
| 09:22 | Ares (local) | git rev-parse, git submodule status, ls                               | Establish repo root, branch, submodules              | master; Home-Lab on audit branch; 8 submodules                      |
| 09:22 | Ares (local) | date, hostname, uptime, os-release                                    | Local host identity                                  | ares, Debian 12, up 15h                                             |
| 09:23 | Ares (local) | read .claude memory + MEMORY.md                                       | Load operational memory                              | 11 memories read                                                    |
| 09:23 | Ares (local) | read Home-Lab CLAUDE.md, audit/                                       | Guidance + prior audit basis                         | audit commit 65f3681                                                |
| 09:24 | 11 hosts     | ping -c1 sweep                                                        | Reachability                                         | all 11 UP (incl. VLAN 30)                                           |
| 09:27 | pve3         | pvecm status, corosync-cfgtool -s                                     | Quorum + corosync ring                               | 7/7 quorate, all links connected                                    |
| 09:27 | QuarkyLab    | nvidia-smi, zpool, qm list, sinfo                                     | GPU, ZFS, guests, SLURM                              | RTX 8000 46GB; workspace ONLINE; VM104; student+research partitions |
| 09:27 | Jarvis       | nvidia-smi, zpool, systemctl is-active                                | GPU, ZFS, AI services                                | 2x RTX 6000; tank+scratch ONLINE; llm_router+ollama active          |
| 09:27 | Randy        | zpool list, zpool status -x/bulk                                      | ZFS health                                           | bulk mpathv FAULTED; datastore ONLINE                               |
| 09:27 | pve1 (.193)  | pveversion, pct list, apt list                                        | Standalone guests + patch lag                        | PVE 9.1.9; CT103+CT104 running; 142 pending                         |
| 09:28 | pve3         | pvesh /cluster/resources, storage.cfg, ha-manager, pvesr              | Cluster guests, storage, HA, replication             | 14 guests; 2 PBS targets; fencing standby; no replication           |
| 09:28 | Randy        | proxmox-backup-manager, zpool status datastore, multipath -ll         | PBS + datastore geometry + drive id                  | datastore 4x RAIDZ2; multipath -ll returned empty                   |
| 09:28 | EX3400       | ssh ex3400 (show version)                                             | Switch state                                         | DENIED (publickey) - confirms R-09                                  |
| 09:30 | pve3         | upsc tripp-lite, upsc midatlantic                                     | UPS state                                            | UPS-A 59% / ~9.5 min; UPS-B 36% / ~29 min                           |
| 09:30 | Randy        | ls -dt /datastore/..., df -h                                          | Backup freshness                                     | 14 snapshots dated 2026-07-23; 97G / 23T                            |
| 18:40 | EX3400       | show version, chassis environment, virtual-chassis, chassis alarms    | Switch identity, power, thermal                      | KM-EX3400; dual PSU OK; 37C; no alarms; VC 1 member                 |
| 18:42 | EX3400       | show spanning-tree bridge/interface, showlldp neighbors               | STP root + neighbors                                 | <b>UniFi was root</b> ; EX3400 at default 32768; cost 20000         |
| 18:44 | EX3400       | show interfaces terse, show interfaces xe-0/2/3                       | Link states                                          | xe-0/2/3 <b>UP 10G</b> ; ge-0/0/45 down, no neighbor                |
| 18:45 | EX3400       | show configuration   display set                                      | Config capture (241 lines)                           | no bridge-priority; native-vlan-id at interface level               |
| 19:05 | Randy        | zfs list -r bulk                                                      | Actual data on bulk                                  | <b>near-empty</b> ; only bulk/backups 57.3G                         |
| 19:10 | 3 ZFS nodes  | zpool status + awk parse validation                                   | Validate new metric logic                            | bulk=1, datastore=0                                                 |
| 21:30 | pve4 CT103   | Prometheus query + max_over_time 7d                                   | Prove alert could not fire                           | pool health max = 0 over 7 days                                     |
| 20:15 | 3 ZFS nodes  | deploy collector + systemctl start; add Grafana rule                  | <b>Modified: Yes</b> - close device-level blind spot | Applied by operator; metric live, bulk=1                            |
| 22:00 | EX3400       | set protocols rstp bridge-priority 4096; commit confirmed then commit | <b>Modified: Yes</b> - pin core as STP root          | Applied by operator; first attempt auto-reverted                    |
| 22:05 | EX3400       | show spanning-tree bridge (post-change)                               | Verify remediation                                   | Root ID == Bridge ID == 4096; ge-0/0/46 ROOT to DESG                |

Two changes were applied during this engagement, both only after explicit operator approval and both verified afterwards: the ZFS device-level metric and alert rule, and the EX3400 spanning-tree bridge-priority. Every other proposed action (Section 20) remains outstanding, including the bulk drive replacement, which is pending operator hardware. Note that the reviewer's own attempts to push these changes directly were blocked by a safety classifier; the operator applied them, which preserved the human-in-the-loop boundary this review recommends.

## B Asset Register

Physical and virtual-host inventory, live-confirmed 2026-07-23 (evidence E-01 through E-06). Full field set is in `data/asset-register.csv`.

| ID   | Host         | Role                                  | CPU / GPU                         | IP / VLAN                  | Status and notes                                                                                                                                              |
|------|--------------|---------------------------------------|-----------------------------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| H-01 | pve1         | Standalone Pi-hole / homepage         | Intel (Mac Mini, runs PVE) / none | .10.193 / v1               | up; NOT a cluster member; PVE 9.1.9; 142 pending; CT103 primary Pi-hole + CT104; NO backup job                                                                |
| H-02 | pve2         | OPNsense host + node                  | i7-8700 / none                    | .10.204 / v1               | up; hosts OPNsense VM100 (edge SPOF); reboot-required; excluded from hardening                                                                                |
| H-03 | pve3         | Core services + rke2-cp1              | i7-8700 / none                    | .10.201 / v1               | up; NPM/Vaultwarden/Homepage/OpenWebUI + VM201; only trunked EliteDesk; e1000e TSO/GSO-off mitigation                                                         |
| H-04 | pve4         | Grafana stack + rke2-cp2              | i5-7500T / none                   | .10.202 / v1               | up; CT103 Grafana/Prom/Loki/InfluxDB + VM202; sustained load ~5                                                                                               |
| H-05 | pve5         | Headscale, Pi-hole2, rke2-cp3, HAOS   | i5-7500T / none                   | .10.203 / v1               | up; CT105 + CT108 + VM110 + VM203                                                                                                                             |
| H-06 | QuarkyLab    | ML / DUNE / student GPU               | 2x E5-2699 v4 / RTX 8000 48GB     | .10.179 (+.30.179) / v1+30 | up; kernel+nvidia-* held (6.14.11-9-pve); workspace raidz1 10.9T; VM104 Wazuh; SLURM                                                                          |
| H-07 | Jarvis       | LLM inference                         | 2x E5-2687W v4 / 2x RTX 6000 24GB | .10.31 (+.30.31) / v1+30   | up; kernel held (nvidia not held - narrower); tank 9.09T + scratch; llm_router+ollama                                                                         |
| H-08 | Randy        | Storage / PBS / Jellyfin / k8s worker | 2x E5-2690 v3 / RX 580            | .10.187 (+.30.187) / v1+30 | up; datastore 36.7T ONLINE; bulk 80T DEGRADED (mpathv); PBS; live PVE 9.2.3 vs docs 9.1.1 (DR-10)                                                             |
| H-09 | DS4246       | JBOD disk shelf                       | NetApp 24-bay / n/a               | dual SAS to Randy          | degraded; 22x 4TB (2 free bays); slot 15 dead 2026-07-21; single enclosure SPOF                                                                               |
| N-01 | EX3400-48P   | Core switch                           | JunOS 23.4R2-S7.4                 | .10.50 / all               | up; KM-EX3400; RSTP root confirmed after bridge-priority 4096 applied 2026-07-23 (QL-R23); dual PSU OK; no alarms; SNMP RO locked to .183/32; xe-0/2/3 UP 10G |
| N-02 | UniFi USW-24 | Access switch                         | UniFi                             | trunk ge-0/0/46            | up                                                                                                                                                            |
| N-03 | UniFi UDR    | Wireless controller / AP              | UniFi                             | .10.2 / v1                 | up; NOT the WAN edge (common misread)                                                                                                                         |
| N-05 | APC AP7901   | Managed PDU                           | APC                               | ge-0/0/38 / v1             | up                                                                                                                                                            |
| P-01 | UPS-B        | Middle Atlantic OL2200R               | feeds R730s+Randy+DS4246          | n/a                        | up; OL, 100%, load 36%, ~29 min                                                                                                                               |
| P-02 | UPS-A        | Tripp Lite SMART1500                  | feeds switch+UniFi+EliteDesks     | n/a                        | up; OL, 100%, load 59%, ~9.5 min (imbalanced)                                                                                                                 |
| W-01 | Ares         | Admin workstation                     | Intel laptop / Debian 12          | .10.199 (+.20.199) / v1+20 | up; ansible/backup/dashboard control node; timers; hourly .claude backup                                                                                      |

## C Service Catalog



Service inventory with placement, exposure, and backup posture. Full field set (auth, TLS, monitoring source, recovery doc) is in data/service-catalog.csv.

| ID   | Service             | Host / guest         | IP : port                       | External                     | Backup / state                              |
|------|---------------------|----------------------|---------------------------------|------------------------------|---------------------------------------------|
| S-01 | OPNsense            | pve2 / VM100         | .10.1 : 443/53                  | LAN edge (dual-WAN)          | randy-pbs 03:00 / running                   |
| S-02 | Pi-hole primary     | pve1 / CT103         | .10.177 : 53/80                 | no                           | NONE (pve1 unbacked) / running              |
| S-03 | Pi-hole secondary   | pve5 / CT108         | .10.178 : 53/80                 | no                           | randy-pbs / running                         |
| S-04 | Unbound             | pve2 / VM100         | .10.1 : 53                      | no                           | with OPNsense / running                     |
| S-05 | Headscale           | pve5 / CT105         | .10.186 : 8080                  | internal                     | randy-pbs / running                         |
| S-06 | Nginx Proxy Manager | pve3 / CT101         | .10.181 : 80/443/81             | yes (proxy front)            | randy-pbs / running                         |
| S-07 | Vaultwarden         | pve3 / CT102         | .30.182 : 80                    | yes (vault.kylemason.org)    | randy-pbs / running                         |
| S-08 | Homepage            | pve3 / CT106         | .10.148 : 3000                  | yes (homepage.kylemason.org) | randy-pbs / running                         |
| S-09 | Homepage (dead)     | pve1 / CT104         | -                               | no                           | NONE / running (should be off, R-12)        |
| S-10 | Open WebUI          | pve3 / CT107         | .30.185 : 8080                  | yes (chat.netframe.local)    | randy-pbs / running                         |
| S-11 | Grafana stack       | pve4 / CT103         | .10.183 : 3000                  | yes (grafana.kylemason.org)  | randy-pbs / running                         |
| S-12 | Prometheus          | pve4 / CT103         | .10.183 : 9090                  | no                           | with Grafana / running                      |
| S-13 | Loki                | pve4 / CT103         | .10.183 : 3100                  | no                           | with Grafana / running (auth_enabled=false) |
| S-14 | Scrutiny            | pve4 / CT103         | .10.183 : 8080                  | no                           | with Grafana / running                      |
| S-15 | Wazuh               | QuarkyLab / VM104    | .10.184 : 443/9200/55000        | yes (wazuh.kylemason.org)    | randy-pbs-10g / running                     |
| S-16 | step-ca             | pve2                 | .10.204 : 443                   | no                           | with pve2 / active                          |
| S-17 | PBS                 | Randy                | .10.187 : 8007                  | no                           | GC+verify weekly / active (single target)   |
| S-18 | Jellyfin            | Randy                | .10.187 : 8096                  | no                           | - / active                                  |
| S-19 | Ollama              | Jarvis               | .10.31 : 11434                  | LAN (unauth, AI-01)          | GPU exporter / active                       |
| S-20 | llm_router          | Jarvis               | .10.31 : 8000                   | yes (llm.netframe.local)     | scripts in git / active (iptables-locked)   |
| S-21 | netframe-monitor    | Jarvis               | .10.31 : 8808/8809              | internal                     | scripts in git / active (15 min)            |
| S-22 | Home Assistant      | pve5 / VM110         | .10.60 : 8123                   | internal                     | randy-pbs 03:30 / running                   |
| S-23 | RKE2                | pve3/4/5 / VM201-203 | .10.51-53 (VIP .54) : 6443/9345 | no                           | randy-pbs 04:00 / running                   |
| S-24 | Private registry    | RKE2                 | registry.netframe.local : 443   | no                           | in-cluster / running (anon, K8S-01)         |
| S-25 | Cloudflare Tunnel   | QuarkyLab            | quarkylab.kylemason.org         | yes (no inbound ports)       | - / active                                  |
| S-26 | SLURM               | QuarkyLab            | .10.179                         | no                           | - / active (shard+MPS+aptainer)             |

## D Port and VLAN Register

VLAN plan and EX3400 core-switch port assignment. The switch was live-inspected on 2026-07-23 (E-22, E-25), so the port map below is verified against running configuration rather than reconstructed. Key-based SSH is still not configured and nightly automated capture is still absent (QL-R16, QL-REC09).

### D.1 VLANs

| ID | Name           | Subnet          | Purpose                                                                                  |
|----|----------------|-----------------|------------------------------------------------------------------------------------------|
| 1  | mgmt / default | 192.168.10.0/24 | Proxmox hosts, corosync, services, switch management (dense)                             |
| 20 | trusted / OOB  | 192.168.20.0/24 | iDRAC x2 + IPMI (BMCs) + Ares jump leg; inbound clamped to Ares; egress-clamped          |
| 30 | servers        | 192.168.30.0/24 | NFS / PBS / GPU egress; dual-homed GPU and storage nodes; blocked to VLAN 1 except :9100 |
| 40 | IoT            | 192.168.40.0/24 | smart-home, IMUs; isolated, DNS pinhole only                                             |
| 50 | VoIP           | 192.168.50.0/24 | CP-8841 phones, FreePBX (deferred)                                                       |

| ID | Name  | Subnet          | Purpose                                      |
|----|-------|-----------------|----------------------------------------------|
| 60 | guest | 192.168.60.0/24 | isolated guest WiFi                          |
| 70 | lab   | 192.168.70.0/24 | CCNA / experimental; standalone sandbox node |

## D.2 EX3400-48P port map

| Port                | Attachment                     | Mode          | VLANs                                                      |
|---------------------|--------------------------------|---------------|------------------------------------------------------------|
| ge-0/0/22           | Jarvis nic2 (onboard 1G)       | Trunk         | native 1 (+ tagged 30, now vestigial)                      |
| ge-0/0/24           | QuarkyLab nic2                 | Trunk         | native 1 + tagged 30                                       |
| ge-0/0/30, /32, /44 | iDRAC / IPMI OOB               | Access        | 20 only                                                    |
| ge-0/0/38           | APC AP7901 PDU                 | Access        | 1                                                          |
| ge-0/0/41           | Ares mgmt / jump               | Dual          | native 1 + tagged 20                                       |
| ge-0/0/45           | unused (no LLDP neighbor)      | Down          | none; the EX2300 never existed (DR-15)                     |
| ge-0/0/46           | UniFi USW-24 Port 24           | Trunk         | native 1 + tagged 20/30/40/50/60/70                        |
| xe-0/2/0            | Randy nic3 (10G ConnectX-3)    | Trunk         | 1 + 30 (storage / NFS)                                     |
| xe-0/2/2            | Jarvis enp132s0 (10G ConnectX) | Access        | 30 (dedicated storage / egress)                            |
| xe-0/2/3            | UniFi SFP2                     | Up 10G (DESG) | native-vlan-id 1; live-confirmed up, DR-14 fault was stale |

Note: native-vlan-id is set at the physical-interface level on this ELS switch, and it IS supported (contrary to a stale note in several docs, D-08). RSTP root is the EX3400 (bridge priority 4096).

## E Backup Matrix

Per-asset backup posture. Last-success dates are live-confirmed for 2026-07-23 (14 snapshots present, evidence E-08). Full field set (encryption, RTO/RPO, failure domain) is in data/backup-matrix.csv. The two rows flagged **HIGH** are the material gaps.

| Asset               | Target                  | Schedule      | Retention  | Last ok | Gap                                                            |
|---------------------|-------------------------|---------------|------------|---------|----------------------------------------------------------------|
| OPNsense VM100      | randy-pbs + config repo | 03:00 + 03:17 | 7d/4w      | 07-23   | VM image only on Randy; config also age-encrypted to GitHub    |
| CT101 NPM           | randy-pbs               | 02:00         | 7d/4w      | 07-23   | single copy                                                    |
| CT102 Vaultwarden   | randy-pbs               | 02:00         | 7d/4w      | 07-23   | secrets single-copy; break-glass offline mitigates             |
| CT103 Grafana       | randy-pbs               | 02:00         | 7d/4w      | 07-23   | single copy                                                    |
| CT105 Headscale     | randy-pbs               | 02:00         | 7d/4w      | 07-23   | single copy                                                    |
| CT106 Homepage      | randy-pbs               | 02:00         | 7d/4w      | 07-23   | single copy                                                    |
| CT107 OpenWebUI     | randy-pbs               | 02:00         | 7d/4w      | 07-23   | single copy                                                    |
| CT108 pi-hole2      | randy-pbs               | 02:00         | 7d/4w      | 07-23   | mirrors CT103 config                                           |
| VM104 Wazuh         | randy-pbs-10g           | 03:00         | 7d/4w      | 07-23   | single copy                                                    |
| VM110 HomeAssistant | randy-pbs               | 03:30         | 7d/4w      | 07-23   | single copy                                                    |
| VM201/202/203 RKE2  | randy-pbs               | 04:00         | 7d/4w      | 07-23   | etcd not separately backed up                                  |
| QuarkyLab workspace | PBS host backup         | 05:30         | per-job    | 07-23   | fernanda refreservation floor                                  |
| CT103/CT104 pve1    | NONE                    | none          | none       | NEVER   | <b>HIGH</b> NO BACKUP; config rebuildable from CT108 (R-03)    |
| bulk pool           | NONE                    | n/a           | n/a        | NEVER   | <b>HIGH</b> NO SECOND COPY of media + fernanda research (R-04) |
| Ares .claude        | randy:/datastore        | hourly        | accumulate | 07-23   | secrets excluded                                               |
| Ares /home          | restic to randy         | 04:03         | restic     | 07-23   | lands ON the degraded bulk pool                                |

All production backups currently reside only on Randy (one PBS node, one rack, one UPS bus): effectively 1-1-1, not 3-2-1 (QL-R03). The Ares workstation restic target sits on the degraded bulk pool, compounding QL-R01. A second, independent copy is the single highest-value recoverability improvement (QL-REC03).

## F Alert Matrix

Grafana Unified Alerting rules (no Alertmanager; all egress to Discord). Full field set (evidence basis, expected response, runbook, false-positive/negative risk) is in `data/alert-matrix.csv`. The GAP-\* rows are alerts that should exist but do not.

| Alert                     | Source           | Sev  | Threshold                                | Channel        | Disposition                   |
|---------------------------|------------------|------|------------------------------------------|----------------|-------------------------------|
| ClusterNotQuorate         | corosync logfile | crit | quorate <1, 2m                           | discord-alerts | Keep                          |
| ClusterVotesBelowExpected | corosync logfile | warn | missing votes >0, 5m                     | discord-alerts | Keep                          |
| InstanceDown              | node / blackbox  | crit | up <1, 2m                                | discord-alerts | Keep                          |
| PiholePrimaryDown         | blackbox DNS     | crit | probe .177 <1, 2m                        | discord-alerts | Keep                          |
| PiholeSecondaryDown       | blackbox DNS     | warn | probe .178 <1, 2m                        | discord-alerts | Keep                          |
| ZfsPoolDegraded           | zpool logfile    | crit | pool_health >0, 5m                       | discord-alerts | Verify delivery (R-02)        |
| GpuTempHigh               | nvidia-smi       | warn | temp >87C, 5m                            | discord-alerts | Keep                          |
| GpuMemoryHigh             | nvidia-smi       | warn | VRAM % >90, 10m                          | discord-alerts | Keep                          |
| DiskAlmostFull            | node_filesystem  | warn | root % >90, 10m                          | discord-alerts | Extend to pools               |
| LowMemory                 | node_memory      | warn | MemAvail % <8, 10m                       | discord-alerts | Keep                          |
| RegistryCertExpiringSoon  | blackbox tls     | warn | cert <12h, 15m                           | discord-alerts | Broaden to all certs          |
| BackupVerifyReportStale   | Randy logfile    | warn | age >26h, 15m                            | discord-alerts | Keep (caught mpathv)          |
| BackupVerifyFailing       | Randy logfile    | warn | overall_pass <1, 5m                      | discord-alerts | Consider critical             |
| AresBackupStale           | Randy logfile    | warn | age >26h, 15m                            | discord-alerts | Keep                          |
| NetframeMonitorStale      | Jarvis heartbeat | warn | age >1h, dead-man                        | discord-alerts | Keep                          |
| UPS battery low           | PeaNUT           | crit | charge <50, 2m                           | discord-ups    | Keep                          |
| UPS runtime low           | PeaNUT           | crit | runtime <300s, 1m                        | discord-ups    | Keep                          |
| PrimaryWANDown            | pve2 logfile     | crit | WAN_GW <1, 1m                            | discord-alerts | Keep                          |
| FirewallBlockRateSpike    | Loki filterlog   | warn | >500 blocks/5m                           | discord-alerts | Keep                          |
| NetworkDeviceConfigChange | loki syslog      | warn | commit >0                                | discord-alerts | Keep                          |
| NetworkSyslogSilent       | Loki stream      | warn | <50 lines/15m, dead-man                  | discord-alerts | Keep                          |
| GAP-quorum-guest          | (none)           | -    | per-guest down (needs pve-exporter)      | -              | Missing (REC06/07)            |
| GAP-pbs-verify-stale      | (none)           | -    | PBS verify job failed / stale            | -              | Missing (REC07)               |
| GAP-host-auth-failure     | (none)           | -    | repeated auth failures per host          | -              | Missing (DETECT-01c)          |
| GAP-cert-expiry-broad     | (none)           | -    | any netframe.local / LE cert near expiry | -              | Missing (only registry today) |
| GAP-2nd-channel           | (none)           | -    | alert egress independent of WAN/Discord  | -              | Missing (REC14)               |

## G Dependency Register

Critical-service dependencies and the impact of losing each dependency. Derived from the service catalog (Appendix C) and `diagrams/service-dependencies.mmd`.

| Service                   | Depends on                         | Impact if dependency lost                                     | Mitigation                                                                    |
|---------------------------|------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------------------------|
| All LAN clients           | OPNsense (VM100, pve2)             | No routing, DHCP, or inter-VLAN; whole network down           | Serial-console DR; nightly age config backup; CARP HA planned (QL-R04)        |
| All name resolution       | Pi-hole .177/.178 to Unbound on .1 | Clients lose DNS; single Unbound is the recursion choke point | Dual Pi-hole HA (DHCP hands out both); Unbound still single (enterprise R-03) |
| All backups, NFS, k8s PVs | Randy (storage / PBS)              | Simultaneous loss of backups, research NFS, and k8s storage   | ZFS raidz2; UPS; second target planned (QL-R05 / QL-REC03)                    |
| Cluster control plane     | corosync quorum (4/7)              | Loss of quorum freezes cluster config and HA                  | 7 votes, healthy; corosync on stable VLAN 1 L2 (E-01)                         |
| Everything                | UPS-A and UPS-B                    | Unclean shutdown; data-at-risk on abrupt power loss           | NUT monitored; verify graceful-shutdown coverage (QL-R17 / R-07)              |
| Full secret set           | Vaultwarden (CT102, in-cluster)    | Cluster-loss recovery cannot read the secrets it needs        | Break-glass age snapshot on Ares + Randy, offline (QL-R08)                    |
| Metrics, logs, alerting   | Grafana stack (CT103, pve4)        | Blind to all metrics/logs and no alert evaluation             | Dead-man heartbeat can catch total loss; single-CT SPOF (QL-R10)              |
| Alert delivery            | WAN + Discord webhook              | No alerts reach the operator during a WAN/Discord outage      | Dead-man alerts; add 2nd channel (QL-R09 / QL-REC14)                          |

| Service                       | Depends on               | Impact if dependency lost                          | Mitigation                                                                      |
|-------------------------------|--------------------------|----------------------------------------------------|---------------------------------------------------------------------------------|
| LLM chat and RAG              | Ollama on Jarvis GPUs    | Chat/RAG down; local inference unavailable         | llm_router auto-fallback to Claude API; deterministic monitoring unaffected     |
| Research and student jobs     | SLURM + Randy NFS + GPU  | No new jobs; running jobs may fail on NFS loss     | Local base.sif copy; refreservation floor; preemption (QL-R02 data single-copy) |
| Kubernetes workloads          | RKE2 etcd + Randy NFS SC | Control plane or PV loss stops workloads           | 3-node etcd quorum; VM-level backups; no etcd-native backup (gap)               |
| Internal TLS (netframe.local) | step-ca (pve2)           | Certs cannot renew; registry / internal TLS breaks | RegistryCertExpiringSoon alert; CA password in Vaultwarden                      |